

CANONICAL DYNAMICAL GEOMETRY SYSTEM
VERSION 4 — INFORMATION-GEOMETRIC FLOW
FRACTAL • CONTACT • GEOMETRIC FLOW • FISHER

ABSTRACT. We define a single, locked ordinary differential equation (the *canonical ODE*) that drives a state $X \in \mathbb{R}^n$ towards a target set encoded by a feature map $S : \mathbb{R}^n \rightarrow \mathbb{R}^k$ and a quadratic energy $E = S^T G S$, while simultaneously executing a domain-specific nominal field F_{base} . Geometry enters the dynamics *only* through the Euclidean gradient $g_X = 2J^T G S$; all projections are Euclidean; the pulled-back metric $J^T G J$ is reserved for curvature analysis. Under standard regularity (Assumptions A1–A4), Part I proves (i) local existence and uniqueness, (ii) Lyapunov descent, (iii) asymptotic stability of $\{E = 0\}$, (iv) exponential convergence with rate $\rho = 4\sigma^2 \mu_G^2 / M_G \cdot (1 - \gamma_{\max})$ under a uniform Jacobian rank condition, and (v) ultimate boundedness when $F_{\text{base}} \neq 0$, with explicit admissibility threshold and a local-in-energy capture window. Part II extends the framework to fractal targets (multi-scale mollified distance with Hausdorff-dimension weighting), contact manifolds (Reeb-aware feature map with horizontal-descent property), and time-varying metrics evolved by Ricci-type geometric flows (strict Lyapunov monotonicity for constant target metrics and practical state–metric bounds in the state-dependent case). Part III specialises the time-varying metric to feature-space Fisher information and its pullback to state space, yielding adaptive preconditioning, a chart-covariant auxiliary natural-gradient flow, and a statistical Cramér–Rao floor on the descent rate. Throughout, the canonical ODE, the gradient formula, and the Euclidean projection are unchanged; all domain content is confined to the triple (S, G, F_{base}) .

CONTENTS

1	Introduction	4
2	Foundations and Assumptions	4
2.1	Standing assumptions	4
2.2	The singular set and its resolution	4
2.3	Well-posedness	5
3	Canonical Core Equation	5
4	Locked Definitions	6
4.1	Representation map	6
4.2	Energy functional	6
4.3	State-space gradient	6
4.4	Modified dynamics	6
4.5	Adaptive control gain	6
5	System Rules (Inviolable)	7
6	Forward Chain	7
§B.2	— State-Space Dynamics	8
§B.2a	— Modified Force Definition	8
§C.5	— Hessian and Curvature Tensor	8
§D.3	— Curvature Manifold	9

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§H.1 — Numerical Implementation	10
7 Main Stability Theorems	11
7.1 Lyapunov descent	11
7.2 Asymptotic stability	11
7.3 Exponential rate	12
7.4 Ultimate boundedness with non-zero F_{base}	12
8 Application Domains	12
8.1 Trading / statistical-arbitrage spread control	12
8.2 Spectral / Galerkin PDE control	13
8.3 First-order optimisation	13
8.4 Task-space robotics control	13
9 Domain Parameterisation	14
10 Verification Checklist	14
11 Structure Freeze	15
12 Summary — One Line	15
A Symbol Table	16
B Quick Reference Card	16
C Proof Roadmap (Next Steps)	16
Part 1. Geometric Extensions	17
D Fractal-Geometric Extension	17
D.1 Motivation and setup	17
D.2 Multi-scale representation	17
D.3 Closure under the canonical core	17
E Contact-Geometric Extension	18
E.1 Motivation	18
E.2 Contact instantiation	18
E.3 Horizontal descent property	18
E.4 Subriemannian special case	19
F Geometric-Flow Extension	19
F.1 Time-varying metric	19
F.2 Modified energy derivative	19
F.3 Joint stability	19
F.4 Discretisation	20
G Unified View of the Three Extensions	20
Part 2. Information-Geometric Flow	20
H Statistical setup and the Fisher metric	20
H.1 Features as statistical parameters	20
H.2 Fisher metric in the canonical system	21
I Natural gradient and chart covariance	21
J Joint stability under Fisher flow	22

K	Three new computational capabilities	22
K.1	Adaptive preconditioning and natural-gradient variant	22
K.2	Variational inference and MLE in one map	22
K.3	Information-geometric Hessian	23
L	Bridge to v5: stochastic extension	23
M	Master unified view (v2 + v3 + v4)	23
Part 3.	Stochastic Canonical SDE	24
N	Canonical SDE	24
N.1	Drift, diffusion, and the canonical SDE	24
N.2	Well-posedness	24
N.3	Itô formula for the energy	25
N.4	The chart-intrinsic noise floor	25
N.5	Mean-square ultimate boundedness	25
N.6	Natural-gradient Langevin and the Gibbs measure	25
N.7	Bridge back to v4	26
N.8	Empirical validation: BTC/ETH/SOL, 2021–2026	26
O	Layered variable glossary	28
O.1	Layer 1 — Core state and geometry	28
O.2	Layer 2 — Gradient and force objects	29
O.3	Layer 3 — The canonical ODE	29
O.4	Layer 4 — Curvature and Hessian objects	30
O.5	Layer 5 — Predictive dashboard scalars	30
O.6	Layer 6 — Stability theorem parameters	31
O.7	Layer 7 — Geometric extensions	31
O.8	Layer 8 — Information-geometric extension	33
O.9	Layer 9 — Thermodynamic dictionary	33
O.10	Layer 10 — Game-theoretic variables	34
O.11	Layer 11 — Multi-engine composition	34
O.12	Layer 12 — Gap-closure variables	34
O.13	Dependency graph	36

LOCKED

Execution paradigm (LOCKED): Euclidean execution with geometric encoding. Geometry enters the dynamics *only* through the state-space gradient g_X . All projections are Euclidean. The pulled-back metric $G_{\text{pull}} = J^T G J$ is used *only* for curvature computation.

1 Introduction

Many engineering and scientific tasks reduce to the same structural problem: drive a state X along a nominal trajectory generated by F_{base} while simultaneously enforcing a geometric constraint encoded by an energy $E(X)$. Examples include statistical-arbitrage portfolios (spread $\rightarrow 0$), spectral PDE controllers (modal error $\rightarrow 0$), first-order optimisers ($\nabla f \rightarrow 0$) and task-space robotics (end-effector error $\rightarrow 0$).

Classical solutions diverge: Riemannian gradient flows replace the state-space metric and obscure execution; control-Lyapunov designs often reinvent the projection at each domain; port-Hamiltonian formulations require an energy-shaping co-design. The cost is a proliferation of incompatible formalisms.

Contribution. We commit to a single, frozen ODE (Section 3) and a single gradient (Section 4.3) that respects three inviolable rules (Section 5) and prove its properties *once*. Domain content is confined to the triple (S, G, F_{base}) (Section 9), so each application becomes an instantiation rather than a redesign.

Organisation. Section 2 fixes assumptions and well-posedness. Sections 3–5 state the locked system. Sections 6–6 develop structural results, the curvature manifold, and the numerical implementation. Section 7 proves the main stability theorems. Section 8 instantiates the framework on four domains. Part II (Sections D–F) extends the kernel by fractal, contact, and geometric-flow constructions. Part III (Section H onwards) specialises Part II to the Fisher information metric and introduces the auxiliary natural-gradient flow.

Related work. Lyapunov-based control and CLF theory [1, 2]; geometric control [3]; gradient flows in metric spaces [4]; neural ODEs [5]; port-Hamiltonian systems [6]; adaptive control [7]. Our framework is closest to Sontag’s universal formula but differs in (a) using a fixed Euclidean projection and (b) routing all geometry through a single feature map and metric.

2 Foundations and Assumptions

This section fixes the regularity class, declares the standing assumptions, resolves the apparent singularity in the canonical ODE, and establishes well-posedness. Every later proof depends on the results stated here.

2.1 Standing assumptions

A1 – A4: Standing assumptions

(A1) Regularity of S :

$S \in C^2(\mathbb{R}^n, \mathbb{R}^k)$. The Jacobian $J(X) = \partial S / \partial X$ and the second derivative $\partial_X J$ are locally bounded.

(A2) Metric:

$G \in \mathbb{R}^{k \times k}$ is symmetric positive-definite and *constant* in X and t . We denote $0 < \mu_G \leq M_G$ for its smallest/largest eigenvalues. (*Relaxed in Parts II–III to (A2’):* $G \in C^1(\mathbb{R}^n, \text{Sym}_+^k)$ with uniform bounds $\mu_G I \preceq G(X) \preceq M_G I$. Every result of Part I is stated under (A2); Parts II–III state explicitly when (A2’) replaces (A2).)

(A3) Nominal field:

$F_{\text{base}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is locally Lipschitz.

(A4) Control parameter:

$\theta > 0$ is fixed.

2.2 The singular set and its resolution

The canonical ODE (Section 3) contains the factor $\|g_X\|^{-2}$ and is undefined on the *singular set* $\Sigma = \{X \in \mathbb{R}^n : g_X(X) = 0\}$.

Lemma 2.1 (Structure of Σ). *Under (A1)–(A2),*

$$g_X(X) = 0 \iff J(X)^T G S(X) = 0.$$

In particular:

- (i) *If $\text{rank } J(X) = k$ (full row rank), then $g_X(X) = 0 \iff S(X) = 0 \iff E(X) = 0$.*
- (ii) *Outside the rank-deficient locus $\mathcal{R} = \{X : \text{rank } J(X) < k\}$, Σ coincides with the target set $\{E = 0\}$.*

Proof. $g_X = 2J^T GS$, hence $g_X = 0 \iff J^T GS = 0$. If J has full row rank, J^T is injective on $\mathbb{R}^k$, so $GS = 0$, and since G is invertible, $S = 0$, giving $E = S^T GS = 0$. Outside $\mathcal{R}$ this holds at every point. $\square$

Regularised canonical ODE

$$\dot{X} = F_{\text{base}} - g_X - \gamma \frac{\langle F_{\text{base}} - g_X, g_X \rangle}{\|g_X\|^2 + \varepsilon} g_X, \quad \varepsilon \geq 0.$$

The *exact* canonical ODE corresponds to $\varepsilon = 0$. In implementation a small $\varepsilon > 0$ avoids division by zero; the limit $\varepsilon \rightarrow 0^+$ recovers the canonical equation away from Σ . By Lemma 2.1 the limit is benign whenever J has full row rank along trajectories.

2.3 Well-posedness

Theorem 2.2 (Local existence and uniqueness). *Assume (A1)–(A4) and let $X_0 \notin \Sigma$. There exists $T_* > 0$ and a unique C^1 solution $X : [0, T_*) \rightarrow \mathbb{R}^n$ of the canonical ODE with $X(0) = X_0$. If $\varepsilon > 0$, the same conclusion holds for every $X_0 \in \mathbb{R}^n$.*

Proof. Let $H(X) = F_{\text{base}}(X) - g_X(X) - \gamma(X) \alpha(X) g_X(X)$ with $\alpha(X) = \langle F_{\text{base}} - g_X, g_X \rangle / (\|g_X\|^2 + \varepsilon)$. Under (A1)–(A3) the maps $X \mapsto S, J, GS, g_X, E, \gamma$ are locally Lipschitz. On $\mathbb{R}^n \setminus \Sigma$ (or all of $\mathbb{R}^n$ if $\varepsilon > 0$), the denominator is bounded away from zero on a neighbourhood of X_0 , so α is locally Lipschitz, and so is H . Picard–Lindelöf yields the conclusion. $\square$

Corollary 2.3 (Global existence under energy bound). *If $\dot{E} \leq 0$ along the solution and the sublevel set $\{E \leq E(X_0)\}$ is compact, the solution is defined for all $t \geq 0$.*

3 Canonical Core Equation

LOCKED

THE CANONICAL ODE (single, unique, not to be modified):

$$\dot{X} = F_{\text{base}}(X) - g_X(X) - \gamma(X) \frac{\langle F_{\text{base}}(X) - g_X(X), g_X(X) \rangle}{\|g_X(X)\|^2} g_X(X)$$

The three terms decompose cleanly:

- $F_{\text{base}}(X)$: domain-specific nominal vector field.
- $-g_X(X)$: geometric correction forcing E to decrease.
- $-\gamma(X) (\cdots) g_X(X)$: adaptive damping that removes the component of $F = F_{\text{base}} - g_X$ aligned with g_X , controlled by energy level.

4 Locked Definitions

All definitions below are *canonical and final*. No variants, no alternatives, no extensions are permitted inside the core system.

4.1 Representation map

A.1 — Representation

$$S : X \mapsto \mathbb{R}^k$$

$X \in \mathbb{R}^n$ is the state. $S(X) \in \mathbb{R}^k$ is the geometric feature vector. $k \geq 1$ is fixed for each domain.

The Jacobian of S at X is

$$J(X) = \frac{\partial S}{\partial X} \in \mathbb{R}^{k \times n}.$$

4.2 Energy functional

A.2 — Energy

$$E(X) = S(X)^T G S(X), \quad G \succ 0, G \in \mathbb{R}^{k \times k}$$

G is symmetric positive-definite. $E(X) \geq 0$ with $E(X) = 0 \iff G^{1/2} S(X) = 0$.

4.3 State-space gradient

A.3 — Gradient (single, unique)

$$g_X(X) = J(X)^T \nabla_S E = 2 J(X)^T G S(X)$$

Derivation. By the chain rule, $\nabla_X E = J^T \nabla_S E$. Since $E = S^T G S$, one has $\nabla_S E = 2GS$. Therefore $g_X = J^T(2GS) = 2J^T GS$. $\square$

No other gradient definition exists in this system.

4.4 Modified dynamics

A.4 — Modified force

$$F(X) = F_{\text{base}}(X) - g_X(X)$$

F is the vector field *before* the adaptive projection term. The full dynamics (Section 3) add the projection of F onto $-g_X$.

4.5 Adaptive control gain

A.5 — Control gain

$$\gamma(X) = \frac{E(X)}{E(X) + \theta}, \quad \theta > 0$$

- $\gamma \in [0, 1)$ for all X .

- $\gamma \rightarrow 0$ as $E \rightarrow 0$: no damping near the target manifold.
- $\gamma \rightarrow 1$ as $E \rightarrow \infty$: full projection damping far away.
- θ is the only tunable scalar in the core system.

5 System Rules (Inviolable)

Rule 1 — Geometry location

Geometry enters *only* through g_X

The matrix G and the map S encode all geometric structure. Nothing else in the core ODE carries geometric information.

Rule 2 — Euclidean projection

All projections use the standard Euclidean inner product on $\mathbb{R}^n$

$$\langle u, v \rangle = u^T v, \quad \|u\| = \sqrt{u^T u}$$

There is no metric-weighted inner product in the dynamics layer.

Rule 3 — Pulled-back metric scope

$G_{\text{pull}} = J^T G J$ is used *only* for curvature (Hessian) analysis

G_{pull} does **not** appear in $\dot{X}$, in g_X , or in any projection.

Rule 4 – No duplication

Each object has exactly one definition; no parallel construction is permitted

In particular: there is no separate $\nabla_X \Phi$, no second force definition, and no alternative projection. The objects S , E , g_X , F , γ and $\dot{X}$ are defined exactly once, in Section 4, and used verbatim everywhere else.

Forbidden

The following are **permanently forbidden** in any version of this document:

- G_{pull}^{-1} appearing in the dynamics.
- Metric-weighted projections ($\langle u, v \rangle_G$ in the ODE).
- An independent $\nabla_X \Phi$ defined separately from g_X .
- A second or alternative definition of the force.
- Duplicate section headers or symbol re-definitions.

6 Forward Chain

The unique, linear data-flow through the system:

$$X \xrightarrow{S} S \xrightarrow{E=S^T G S} E \xrightarrow{2J^T G S} g_X \xrightarrow{\dot{X}=F_{\text{base}}-g_X-\gamma(\cdots)g_X} \dot{X}$$

Each arrow is a unique map; no step branches or has an alternative.

§B.2 State-Space Dynamics

Setup. All dynamical quantities live in the state space $\mathbb{R}^n$. The representation map S and the energy E are scalar objects computed from X but the forces F_{base} , g_X , and F are all $\mathbb{R}^n$ -valued.

Projection. The projection of any vector $v \in \mathbb{R}^n$ onto g_X is

$$\text{proj}_{g_X}(v) = \frac{\langle v, g_X \rangle}{\|g_X\|^2} g_X,$$

computed with the *Euclidean* inner product. The complement is $v - \text{proj}_{g_X}(v)$.

Geometric content. The geometry of the target manifold/set is entirely captured by g_X . Equations of motion do not reference G , J , or S directly; they see only g_X computed from those quantities.

§B.2a Modified Force Definition

§B.2a — Modified Force (authoritative, single definition)

$$F(X) = F_{\text{base}}(X) - g_X(X)$$

The full canonical ODE (Section 3) is equivalently written as:

$$\dot{X} = F(X) - \gamma(X) \frac{\langle F(X), g_X(X) \rangle}{\|g_X(X)\|^2} g_X(X).$$

This removes from F its component along g_X , scaled by γ .

§C.5 Hessian and Curvature Tensor

This section derives the Hessian of E explicitly. The derivation introduces the second-derivative tensor of the representation map, which is the only tensorial object the system requires.

The representation curvature tensor K

C.5.0 — Representation curvature tensor

For $S \in C^2(\mathbb{R}^n, \mathbb{R}^k)$ define

$$K^i_{ab}(X) := \partial_a \partial_b S^i(X) = \partial_a J^i_b(X), \quad i \in \{1, \dots, k\}, \quad a, b \in \{1, \dots, n\}.$$

$K(X) \in \mathbb{R}^{k \times n \times n}$ is the third-order tensor of second partials of S . By Schwarz's theorem $K^i_{ab} = K^i_{ba}$; each *slice* $K^i \in \mathbb{R}^{n \times n}$ is symmetric.

Derivation of the Hessian of E

Proposition 6.1 (State-space Hessian, derived form). *Under (A1)–(A2), with index convention as above and Einstein summation,*

$$(\nabla_X^2 E)_{ab} = 2(J^T G J)_{ab} + \eta_i K^i_{ab}, \quad \eta := \nabla_S E = 2GS.$$

Equivalently, in coordinate-free form,

$$\nabla_X^2 E = 2G_{\text{pull}} + \langle \eta, K \rangle_1, \quad \langle \eta, K \rangle_1 := \sum_{i=1}^k \eta_i K^i.$$

Proof. Write $E = G_{ij} S^i S^j$ with $G^T = G$. Then

$$\begin{aligned}\partial_a E &= G_{ij} (\partial_a S^i) S^j + G_{ij} S^i (\partial_a S^j) \\ &= 2G_{ij} J^i_a S^j = 2(J^T G S)_a = (g_X)_a,\end{aligned}$$

recovering $\nabla_X E = g_X$. Differentiating once more,

$$\begin{aligned}\partial_b \partial_a E &= 2G_{ij} ((\partial_b J^i_a) S^j + J^i_a (\partial_b S^j)) \\ &= 2G_{ij} K^i_{ab} S^j + 2G_{ij} J^i_a J^j_b \\ &= \underbrace{(2GS)_i K^i_{ab}}_{= \eta_i K^i_{ab}} + 2(J^T G J)_{ab}.\end{aligned}$$

Symmetry in (a, b) holds because K^i and $J^T G J$ are symmetric. $\square$

Remark 6.2 (Two ingredients). The state-space Hessian decomposes uniquely into:

- the **pulled-back metric** $2G_{\text{pull}} = 2J^T G J$, capturing the linear part of the geometry;
- the **curvature contraction** $\langle \eta, K \rangle_1$, capturing how the representation map S itself bends.

The curvature contraction vanishes whenever S is affine ($K \equiv 0$) or when $\eta = 2GS = 0$ (i.e. on the target set $\{E = 0\}$). Hence on the target set, $\nabla_X^2 E|_{\{E=0\}} = 2G_{\text{pull}}$ exactly.

Remark 6.3 (Operator norm). Define the tensor norm

$$\|K\|_{\text{op}} := \sup_{\|u\|=\|v\|=1, \|w\|=1} \sum_{i,a,b} K^i_{ab} u_i v_a w_b.$$

Then the curvature contraction obeys $\|\langle \eta, K \rangle_1\|_2 \leq \|K\|_{\text{op}} \|\eta\|_2$, which is the bound used in Proposition 6.6.

Remark 6.4 (Role in the dynamics). $G_{\text{pull}} = J^T G J$ and K both appear here as part of the Hessian. By Rule 3, neither appears in $\dot{X}$ or g_X .

§D.3 Curvature Manifold

§D.3 — Curvature manifold

$$\mathcal{C}_{\text{man}} = \{X \in \mathbb{R}^n : \lambda_{\max}(\nabla_X^2 E(X)) = 1\}$$

$\mathcal{C}_{\text{man}}$ is the level set of maximal principal curvature equal to 1. It separates regions where the energy landscape is convex-dominated ($\lambda_{\max} < 1$) from regions where nonlinear curvature terms dominate ($\lambda_{\max} > 1$).

Proposition 6.5. $\mathcal{C}_{\text{man}}$ is generically a smooth $(n-1)$ -dimensional submanifold of $\mathbb{R}^n$ wherever $\lambda_{\max}$ is a simple eigenvalue and its derivative is non-zero.

Proof. Let $h(X) = \lambda_{\max}(\nabla_X^2 E(X)) - 1$ so that $\mathcal{C}_{\text{man}} = h^{-1}(0)$. By (A1) the entries of $\nabla_X^2 E$ are continuous; on the open set where $\lambda_{\max}$ is simple, classical perturbation theory (Kato) gives $\lambda_{\max} \in C^1$ with derivative $\nabla h = v^T (\partial_X \nabla_X^2 E) v$, where v is the unit eigenvector. Wherever $\nabla h \neq 0$ the implicit function theorem yields the smooth codimension-1 structure. $\square$

Proposition 6.6 (Curvature bound). Under (A1)–(A2), with $\|K\|_{\text{op}}$ as defined in Section 6,

$$\lambda_{\max}(\nabla_X^2 E) \leq 2M_G \|J\|_2^2 + \|K\|_{\text{op}} \|\eta\|_2, \quad \eta = 2GS.$$

In particular, on $\mathcal{C}_{\text{man}}$, $2M_G \|J\|_2^2 + \|K\|_{\text{op}} \|2GS\|_2 \geq 1$.

Proof. By Proposition 6.1, $\nabla_X^2 E = 2J^T GJ + \langle \eta, K \rangle_1$. The spectral norm is sub-additive; bound the first term using $\|J^T GJ\|_2 \leq \|G\|_2 \|J\|_2^2 \leq M_G \|J\|_2^2$, and the second using $\|\langle \eta, K \rangle_1\|_2 \leq \|K\|_{\text{op}} \|\eta\|_2$ from Section 6. $\square$

The practical role of $\mathcal{C}_{\text{man}}$:

- Provides a natural domain boundary for local linearisation arguments.
- Guides the choice of θ in γ : setting θ so that $\gamma \approx \frac{1}{2}$ on $\mathcal{C}_{\text{man}}$ balances correction and tracking.

§H.1 Numerical Implementation

H.1.1 Per-step algorithm

Given state X_t , advance one step:

1. Compute $S \leftarrow S(X_t)$.
2. Compute $J \leftarrow \partial S / \partial X|_{X_t}$.
3. Compute $g_X \leftarrow 2J^T GS$. *(gradient — **single computation**)*
4. Compute $E \leftarrow S^T GS$. *(energy)*
5. Compute $F_{\text{base}} \leftarrow F_{\text{base}}(X_t)$.
6. Compute $F \leftarrow F_{\text{base}} - g_X$. *(modified force — **not** F_{base} alone)*
7. Compute $\gamma \leftarrow E / (E + \theta)$.
8. Compute projection coefficient $\alpha \leftarrow \langle F, g_X \rangle / \|g_X\|^2$.
9. Set $\dot{X} \leftarrow F - \gamma \alpha g_X$. *(canonical ODE)*
10. Integrate: $X_{t+h} \leftarrow X_t + h \dot{X}$ (or higher-order scheme).
11. **Termination:** stop when $E < \varepsilon_E$ and $\|g_X\| < \varepsilon_g$ for τ consecutive steps (cf. §H.1.4).

Forbidden

Step 6 uses $F = F_{\text{base}} - g_X$, not F_{base} alone. Step 8 uses this F in the projection, not F_{base} .

H.1.2 Gradient correctness check

Before running any simulation, verify:

$$g_X \stackrel{?}{=} 2J^T GS,$$

by finite-difference: $\Delta E / \Delta X_i$ should match $(g_X)_i$ to machine precision for small perturbations.

H.1.3 Energy monotonicity (derivation)

Lemma 6.7 (Energy derivative). *Along any solution of the canonical ODE,*

$$\dot{E} = \langle g_X, \dot{X} \rangle = (1 - \gamma) (\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2).$$

Equivalently, with $F = F_{\text{base}} - g_X$ and $\alpha = \langle F, g_X \rangle / \|g_X\|^2$,

$$\dot{E} = \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma \alpha \|g_X\|^2 = \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma \langle F, g_X \rangle.$$

Proof. Differentiate $E = S^T GS$: $\dot{E} = 2S^T G \dot{S} = 2S^T G J \dot{X} = (2J^T GS)^T \dot{X} = \langle g_X, \dot{X} \rangle$. Substitute the canonical ODE $\dot{X} = F - \gamma \alpha g_X$ to get $\dot{E} = \langle g_X, F \rangle - \gamma \alpha \|g_X\|^2 = \langle g_X, F \rangle - \gamma \langle F, g_X \rangle = (1 - \gamma) \langle F, g_X \rangle$. Since $\langle F, g_X \rangle = \langle F_{\text{base}}, g_X \rangle - \|g_X\|^2$, the two displayed forms follow. $\square$

Pure geometric flow ($F_{\text{base}} = 0$). Then $F = -g_X$, so

$$\dot{E} = -(1 - \gamma) \|g_X\|^2 \leq 0,$$

strictly negative whenever $g_X \neq 0$ (and $\gamma < 1$, always).

Energy descent condition. Since $1 - \gamma > 0$, $\dot{E} \leq 0$ holds if and only if

$$\boxed{\langle g_X, F_{\text{base}} \rangle \leq \|g_X\|^2.}$$

This is the explicit angle condition between F_{base} and g_X used in the stability theorems of Section 7.

H.1.4 Step-size, stopping, and Jacobian computation

Stopping criterion. Halt when both $E(X_t) < \varepsilon_E$ and $\|g_X(X_t)\| < \varepsilon_g$ hold for τ consecutive steps (dwell time), with default tolerances $\varepsilon_E = 10^{-8}$, $\varepsilon_g = 10^{-6}$, $\tau = 5$.

Adaptive step-size. Use $h_t = h_0 / (1 + \lambda_{\max}(\nabla_X^2 E(X_t)))$, so $h_t \leq h_0$ everywhere and $h_t \rightarrow h_0/2$ on $\mathcal{C}_{\text{man}}$. By Proposition 6.6, $\lambda_{\max}$ is bounded in terms of $\|J\|$, $\|G\|$, and the curvature of S , so h_t is uniformly positive on bounded sets.

Choice of G . The conditioning of $g_X = 2J^T G S$ scales with $\kappa(G)$. Default rule: choose G diagonal with $G_{ii} = 1/\widehat{\text{Var}}(S_i)$ from data (or design), so E becomes the squared Mahalanobis norm of S and the energy landscape is approximately isotropic in the operating region.

Jacobian computation. In order of preference: (i) closed-form analytic J ; (ii) automatic differentiation when S is given as code; (iii) finite differences with step $\delta = \sqrt{\epsilon_{\text{mach}}}$ at cost $O(nk)$ evaluations per step.

7 Main Stability Theorems

We now state the principal results of the paper. Throughout this section (A1)–(A4) hold and the trajectory $X(\cdot)$ is a solution of the canonical ODE on a maximal interval of existence.

7.1 Lyapunov descent

Theorem 7.1 (Lyapunov descent). *For any solution $X(\cdot)$ on $\mathbb{R}^n \setminus \Sigma$,*

$$\dot{E}(t) = (1 - \gamma) (\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2).$$

In particular:

- (i) *If $F_{\text{base}} \equiv 0$, then $\dot{E} = -(1 - \gamma)\|g_X\|^2 < 0$ whenever $g_X \neq 0$.*
- (ii) *In general, $\dot{E} \leq 0$ iff $\langle g_X, F_{\text{base}} \rangle \leq \|g_X\|^2$.*

Proof. Direct from Lemma 6.7; (i) is the special case $F_{\text{base}} = 0$, and (ii) follows since $1 - \gamma > 0$. $\square$

7.2 Asymptotic stability

Theorem 7.2 (Asymptotic stability of the target set). *Assume (A1)–(A4), $F_{\text{base}} \equiv 0$, and that $J(X)$ has full row rank on every sublevel set $\{E \leq c\}$ that is compact. Then for every initial condition X_0 with $\{E \leq E(X_0)\}$ compact, the solution exists for all $t \geq 0$ and*

$$E(X(t)) \xrightarrow[t \rightarrow \infty]{} 0, \quad \text{dist}(X(t), \{E = 0\}) \rightarrow 0.$$

Proof. By Theorem 7.1(i) $\dot{E} \leq 0$, so $X(t)$ stays in the compact sublevel set; Corollary 2.3 gives global existence. LaSalle's invariance principle implies $X(t)$ converges to the largest invariant set inside $\{\dot{E} = 0\} = \{g_X = 0\}$. By Lemma 2.1(i) the rank assumption forces $\{g_X = 0\} = \{E = 0\}$, which is itself invariant (since the right-hand side vanishes there). Hence $E(X(t)) \rightarrow 0$. $\square$

7.3 Exponential rate

Theorem 7.3 (Exponential convergence). *Assume the hypotheses of Theorem 7.2 and that on the invariant compact set there exists $\sigma > 0$ with $\sigma_{\min}(J(X)) \geq \sigma$. Then*

$$E(X(t)) \leq E(X_0) e^{-\rho t}, \quad \rho = 4\sigma^2 \frac{\mu_G^2}{M_G} (1 - \gamma_{\max}),$$

where $\gamma_{\max} = E(X_0)/(E(X_0) + \theta) < 1$.

Proof. With $F_{\text{base}} = 0$, $\dot{E} = -(1 - \gamma)\|g_X\|^2$. We bound $\|g_X\|^2 = 4\|J^T GS\|^2 \geq 4\sigma^2\|GS\|^2 \geq 4\sigma^2\mu_G^2\|S\|^2$, while $E = S^T GS \leq M_G\|S\|^2$, giving $\|S\|^2 \geq E/M_G$. Hence

$$\|g_X\|^2 \geq \frac{4\sigma^2\mu_G^2}{M_G} E.$$

With $\gamma \leq \gamma_{\max} < 1$ monotone in E , $\dot{E} \leq -\rho E$ and Grönwall's inequality concludes. $\square$

7.4 Ultimate boundedness with non-zero F_{base}

Theorem 7.4 (Ultimate boundedness). *Assume (A1)–(A4), the rank condition of Theorem 7.3, and that $\|F_{\text{base}}\| \leq M$ on the operating region. Define*

$$\Psi(R) := \frac{\theta}{R + \theta} \cdot 2\sigma \sqrt{\frac{\mu_G^2}{M_G}} R = (1 - \gamma(R)) \cdot 2\sigma \frac{\mu_G}{\sqrt{M_G}} \sqrt{R}.$$

Then Ψ is unimodal on $(0, \infty)$ with maximum $\Psi^ = \sigma\theta\mu_G/\sqrt{M_G}$ attained at $R^* = \theta$. Assuming the admissibility condition $M < \Psi^*$, the equation $\Psi(R) = M$ has two roots $0 < R_- \leq R_+$. If the trajectory enters the admissible energy window $E(X(t_0)) \leq R_+$, then for any $R \in [R_-, R_+]$, $E(X(t)) \leq R$ for all sufficiently large $t \geq T(R, X(t_0))$. In particular, the theorem is a local-in-energy ultimate-boundedness result on the operating window $\{E \leq R_+\}$; it does not claim global capture from arbitrarily large energies.*

Proof. From Lemma 6.7, $\dot{E} = (1 - \gamma)(\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2) \leq -(1 - \gamma)\|g_X\|(\|g_X\| - M)$. On $\{E \geq R\}$ the rank condition and the bound from Theorem 7.3 give $\|g_X\|^2 \geq 4\sigma^2(\mu_G^2/M_G)R$, so $(1 - \gamma(R))\|g_X\| \geq \Psi(R)$. For $R \in [R_-, R_+]$, $\Psi(r) \geq M$ for every $r \in [R, R_+]$. Therefore $\dot{E} \leq 0$ on the annulus $\{R \leq E \leq R_+\}$, so once a trajectory is in the operating window $\{E \leq R_+\}$ it cannot cross upward through R_+ and is pulled into $\{E \leq R\}$ in finite time. $\square$

Remark 7.5 (Critical disturbance). The admissibility threshold $\Psi^* = \sigma\theta\mu_G/\sqrt{M_G}$ is the largest disturbance amplitude M for which any sublevel set is forward-invariant. For $M \geq \Psi^*$, no finite ultimate bound is provided by the canonical descent alone; an outer feedback shaping F_{base} is required.

8 Application Domains

Each subsection instantiates (S, G, F_{base}) for a specific domain, verifies (A1)–(A4), and cites the appropriate stability theorem from Section 7. *No structural change to the canonical ODE occurs.*

8.1 Trading / statistical-arbitrage spread control

Let $X = w \in \mathbb{R}^n$ be portfolio weights, $A \in \mathbb{R}^{k \times n}$ a factor-loading matrix, $b \in \mathbb{R}^k$ a target spread vector, and $\Sigma \in \mathbb{R}^{k \times k}$ the (PD) factor covariance.

$$S(w) = Aw - b, \quad G = \Sigma^{-1}, \quad F_{\text{base}}(w) = -\kappa(w - w^*),$$

with $\kappa > 0$ and w^* the alpha-target weight. Then $J = A$ (constant), $E(w) = (Aw - b)^T \Sigma^{-1} (Aw - b)$ is the squared Mahalanobis spread, and $g_X = 2A^T \Sigma^{-1} (Aw - b)$.

Corollary 8.1. *If A has full row rank, Theorem 7.3 applies with $\sigma = \sigma_{\min}(A)$ and gives exponential decay of the Mahalanobis spread; Theorem 7.4 bounds the long-run spread in terms of the alpha-tracking torque $\kappa\|w_0 - w^*\|$.*

Transaction costs enter via a Lipschitz extension of F_{base} (smooth proximal regularisation), preserving (A3).

8.2 Spectral / Galerkin PDE control

Let $X = (a_1, \dots, a_N) \in \mathbb{R}^N$ be the Galerkin coefficients of a target field, and a^* a desired spectrum.

$$S(X) = X - a^*, \quad G = \text{diag}(g_1, \dots, g_N), \quad F_{\text{base}}(X) = \mathcal{L}_N X,$$

where $g_k > 0$ are mode weights (e.g. $g_k = k^2$ for an H^1 -type energy) and $\mathcal{L}_N$ is the discretised PDE operator. Then $J = I_N$, $E = \sum_k g_k (a_k - a_k^*)^2$, and $g_X = 2G(X - a^*)$.

Corollary 8.2. *The rank condition is automatic ($J = I_N$). Theorem 7.3 gives convergence rate $\rho = 4\mu_G(1 - \gamma_{\max})$ in the weighted ℓ^2 norm. Stability of the underlying PDE in the full space requires $\sup_N \|\mathcal{L}_N\| < \infty$ in the weighted norm; under that uniform bound, the discrete results lift to the continuum limit.*

8.3 First-order optimisation

Minimise $f \in C^2(\mathbb{R}^n)$ with locally bounded Hessian. Set

$$S(X) = \nabla f(X), \quad G = I_n, \quad F_{\text{base}}(X) = -\nabla f(X).$$

Then $J(X) = \nabla^2 f(X)$, $E(X) = \|\nabla f(X)\|^2$, and $g_X = 2(\nabla^2 f) \nabla f$. The canonical ODE becomes a damped, projection-corrected gradient flow that vanishes at stationary points.

Corollary 8.3. *If $\nabla^2 f$ is uniformly positive-definite on a sublevel set of f , then $\sigma_{\min}(J) \geq \sigma > 0$ on that set, Theorem 7.3 applies, and $\|\nabla f(X(t))\|^2$ decays exponentially.*

Comparison with classical preconditioned descent: the canonical ODE adds the γ -modulated projection term, which bounds the transient overshoot of E at the cost of one extra inner product per step.

8.4 Task-space robotics control

Let $X = q \in \mathbb{R}^n$ be joint angles, $\text{FK} : \mathbb{R}^n \rightarrow \mathbb{R}^k$ the forward kinematics map, $p^* \in \mathbb{R}^k$ the desired end-effector pose, and $W \succ 0$ a task-space weight.

$$S(q) = \text{FK}(q) - p^*, \quad G = W, \quad F_{\text{base}}(q) = -K_p J(q)^T W (\text{FK}(q) - p^*),$$

with $K_p > 0$. Note $F_{\text{base}} = -\frac{K_p}{2}g_X$, so $F = F_{\text{base}} - g_X = -(K_p/2 + 1)g_X$ is fully aligned with $-g_X$, the projection coefficient becomes identically 1, and the canonical ODE collapses to scalar-damped descent $\dot{X} = -(1 + K_p/2)(1 - \gamma)g_X$. This domain therefore illustrates a degenerate alignment regime; richer behaviour arises when F_{base} is replaced by a state-feedback controller not collinear with g_X .

Corollary 8.4. *Away from kinematic singularities ($\text{rank } J = k$) the rank condition holds and Theorem 7.3 gives exponential pose convergence. Near singularities use the regularised ODE ($\varepsilon > 0$); under the regularisation the conclusion of Theorem 7.2 weakens to convergence to the closure of the invariant set $\{S \in \ker J^T\}$.*

9 Domain Parameterisation

LOCKED

Domain independence of the core system:

Different domains change *only* $S(X)$, $E(X)$, $F_{\text{base}}(X)$.

The ODE structure, the gradient formula, the projection, and the control gain are unchanged across all domains.

Domain	$S(X)$	$E(X)$	$F_{\text{base}}(X)$
Trading / Signal	Spread / factor vector	$S^T G S$ with $G = \text{diag}(\sigma^{-2})$	Alpha signal + mean reversion
PDE control	Modal coefficients a_k	$\ a\ _G^2$ (weighted L^2)	PDE right-hand side
Optimisation	Gradient / constraint residual	$\ \nabla f\ ^2$ or barrier term	Gradient descent step
Robotics	Task-space error	$e^T W e$	Joint velocity command

Remark 9.1. The table entries are *examples*, not definitions. Any choice of S , G , and F_{base} satisfying the structural requirements of Sections 4.1–4.5 produces a valid instance of the canonical system.

10 Verification Checklist

All items below must evaluate to **TRUE** before any application or proof proceeds. If any item is FALSE, return to the indicated section and fix it.

Verification checklist

- ✓ **1. Forward chain intact**
 $X \rightarrow S \rightarrow E \rightarrow g_X \rightarrow \dot{X}$ is the unique data path. → §6
- ✓ **2. Single gradient**
 $g_X = 2J^T G S$ is the only gradient appearing anywhere. → §4.3
- ✓ **3. Single force definition**
 $F = F_{\text{base}} - g_X$; no other force is named or used. → §6
- ✓ **4. Euclidean projection only**
 $\langle \cdot, \cdot \rangle = u^T v$; no $\langle \cdot, \cdot \rangle_G$ in dynamics. → Rule 2
- ✓ **5. No metric mixing**
 G_{pull} absent from $\dot{X}$ and g_X . → Rule 3
- ✓ **6. Control uses F , not F_{base}**
Projection term $\langle F, g_X \rangle$; not $\langle F_{\text{base}}, g_X \rangle$. → §3
- ✓ **7. Hessian formula correct**
 $\nabla_X^2 E = 2J^T G J + \langle \eta, K \rangle_1$ with $\eta = 2GS$ and $K^i_{ab} = \partial_a \partial_b S^i$. → §C.5
- ✓ **8. Curvature manifold defined**
 $\mathcal{C}_{\text{man}} = \{X : \lambda_{\max}(\nabla_X^2 E) = 1\}$. → §D.3
- ✓ **9. Domain variation confined**
Only S , E , F_{base} differ across domains. → §9

✓ **10. No structural changes after this version**
Applications, proofs, and implementations only.

→ §11

11 Structure Freeze

LOCKED

NO further structural changes to this system.

The core ODE (§3), the five definitions (§4), and the four rules (§5) are **frozen**.

Permitted after this point:

- **Applications** — instantiate S , G , F_{base} for a new domain.
- **Proofs** — stability, convergence, curvature bounds.
- **Implementations** — numerical integrators, code, experiments.

Not permitted:

- Redefining g_X , F , γ , or the ODE.
- Adding metric-weighted projections.
- Introducing alternative gradient or force terms.

12 Summary — One Line

Lock system → Build applications → Prove results

A Symbol Table

Symbol	Meaning	Dimensions
X	State vector	$\mathbb{R}^n$
$S(X)$	Representation / feature map	$\mathbb{R}^k$
$J(X)$	Jacobian $\partial S / \partial X$	$\mathbb{R}^{k \times n}$
G	Feature-space metric (SPD, fixed)	$\mathbb{R}^{k \times k}$
$E(X)$	Energy = $S^T G S$	$\mathbb{R}_{\geq 0}$
$g_X(X)$	State-space gradient = $2J^T G S$	$\mathbb{R}^n$
$F_{\text{base}}(X)$	Domain-specific nominal field	$\mathbb{R}^n$
$F(X)$	Modified force = $F_{\text{base}} - g_X$	$\mathbb{R}^n$
$\gamma(X)$	Adaptive gain = $E / (E + \theta)$	$[0, 1]$
θ	Gain regularisation constant	$\mathbb{R}_{> 0}$
G_{pull}	Pulled-back metric = $J^T G J$	$\mathbb{R}^{n \times n}$
$K(X)$	Curvature tensor of S , $K^i_{ab} = \partial_a \partial_b S^i$	$\mathbb{R}^{k \times n \times n}$
$\eta(X)$	Feature-gradient = $\nabla_S E = 2GS$	$\mathbb{R}^k$
$\mathcal{C}_{\text{man}}$	Curvature manifold $\lambda_{\max} = 1$	$\subset \mathbb{R}^n$

B Quick Reference Card

Quantity	Formula
Representation	$S : X \rightarrow \mathbb{R}^k$
Energy	$E = S^T G S, \quad G \succ 0$
Gradient	$g_X = 2J^T G S$
Modified force	$F = F_{\text{base}} - g_X$
Gain	$\gamma = E / (E + \theta)$
Canonical ODE	$\dot{X} = F_{\text{base}} - g_X - \gamma \frac{\langle F, g_X \rangle}{\ g_X\ ^2} g_X$
Hessian	$\nabla_X^2 E = 2J^T G J + \langle \eta, K \rangle_1, \quad \eta = 2GS$
Curvature manifold	$\mathcal{C}_{\text{man}} = \{X : \lambda_{\max}(\nabla_X^2 E) = 1\}$

C Proof Roadmap (Next Steps)

The following results remain to be established (proofs are outside the locked core but follow directly from it):

P1. Lyapunov descent. *Done* – Theorem 7.1.

P2. Invariant set / asymptotic stability. *Done* – Theorem 7.2.

- P3. Curvature bound.** *Done* – Proposition 6.6.
- P4. Domain convergence.** *Done* – corollaries in Section 8.
- P5. Discretisation error.** *Open.* Bound integration error growth for Euler and Runge–Kutta schemes as a function of h and θ ; quantify the bias introduced by $\varepsilon > 0$ regularisation.
- P6. Stochastic extension.** *Open.* Itô analysis of $dX = (\cdots) dt + \sigma dW$, including the Itô correction to $\dot{E}$.
- P7. Time-varying $G(t)$.** *Open.* Effect on $\dot{E}$ and on the rate ρ of Theorem 7.3.

Part 1. Geometric Extensions

LOCKED

Extension principle (v3). Each extension below adds new geometric structure *strictly upstream* of g_X , by changing either the representation map S , the metric G , or the energy template. **The canonical ODE, the gradient formula $g_X = 2J^T G S$, and the Euclidean projection are unchanged.** The frozen v2 system from Part I is the *kernel* of every extension.

D Fractal-Geometric Extension

D.1 Motivation and setup

Many target sets are not smooth manifolds but *self-similar* attractors with non-integer Hausdorff dimension $d_H \in (0, n)$: financial support/resistance, turbulent attractors, multi-scale boundary layers. The frozen core requires only that the set be expressed as the zero level set of a smooth energy. We therefore route all fractal information through S .

Let $\mathcal{A} \subset \mathbb{R}^n$ be a (possibly fractal) compact target and fix $\delta_0 > 0$. Define the δ -*mollified distance*

$$d_\delta(X) := \inf_{Y \in \mathcal{A}_\delta} \|X - Y\|, \quad \mathcal{A}_\delta := \{Y : \text{dist}(Y, \mathcal{A}) \leq \delta\}.$$

For $\delta > 0$, $\mathcal{A}_\delta$ is a smooth tubular thickening, so $d_\delta \in C^2(\mathbb{R}^n \setminus \mathcal{A}_\delta)$.

D.2 Multi-scale representation

F.1 — Fractal feature map

Fix scales $\delta_1 > \delta_2 > \cdots > \delta_L > 0$ and define

$$S_{\text{frac}}(X) := (d_{\delta_1}(X), \dots, d_{\delta_L}(X))^T \in \mathbb{R}^L.$$

The fractal metric is $G_{\text{frac}} = \text{diag}(\delta_1^{-2d_H}, \dots, \delta_L^{-2d_H})$.

Why this metric. For a d_H -dimensional self-similar set, $\text{Vol}(\mathcal{A}_\delta) \sim \delta^{n-d_H}$ and the typical squared distance at scale δ scales as δ^2 . The weight $\delta_\ell^{-2d_H}$ rescales the contribution of each shell so that mass is balanced across scales according to the Hausdorff dimension. Strict scale-invariance under simultaneous dilation $\mathcal{A} \mapsto \lambda \mathcal{A}$, $\delta \mapsto \lambda \delta$ (where each summand transforms as λ^{2-2d_H}) holds only at $d_H = 1$; for general d_H the weight produces *dimension-aware* but not invariant weighting, which is the right notion for capacity-balanced descent.

D.3 Closure under the canonical core

Proposition D.1 (Fractal instantiation). *With $S = S_{\text{frac}}$, $G = G_{\text{frac}}$, and locally Lipschitz F_{base} , assumptions (A1)–(A4) hold on the open set $\Omega := \mathbb{R}^n \setminus (\mathcal{A}_{\delta_L} \cup \text{Cut}(\mathcal{A}))$, where $\text{Cut}(\mathcal{A})$ is the medial axis of $\mathcal{A}$. On Ω each d_{δ_ℓ} is C^2 and Theorems 7.1–7.4 apply with the rate constant*

of Theorem 7.3. Globally, $d_{\delta_\ell} \in C^{1,1}$, and (A1) can be replaced by (A1') ($S \in C^{1,1}$ with locally bounded weak Hessian) while preserving Theorems 7.1 and 7.2.

Proof. On Ω the squared-distance functions are C^2 by Federer's tubular-neighbourhood theorem; G_{frac} is constant SPD; apply Theorem 2.2. The $C^{1,1}$ extension follows from Rademacher and standard a.e. chain-rule arguments. $\square$

Proposition D.2 (Multi-scale exponential decay). *For $F_{\text{base}} \equiv 0$ on Ω ,*

$$E_{\text{frac}}(X(t)) \leq E_{\text{frac}}(X_0) e^{-\rho_{\text{frac}} t}, \quad \rho_{\text{frac}} = 4 \sigma_J^2 \frac{\delta_L^{-4d_H}}{\delta_1^{-2d_H}} (1 - \gamma_{\max}),$$

with σ_J the smallest singular value of the Jacobian of S_{frac} on the operating sublevel set.

Proof. Apply Theorem 7.3 with $\mu_G = \delta_L^{-2d_H}$ and $M_G = \delta_1^{-2d_H}$ (eigenvalues of the diagonal G_{frac}). $\square$

E Contact-Geometric Extension

E.1 Motivation

Many controlled systems live on a contact manifold (M^{2m+1}, α) with $\alpha \wedge (d\alpha)^m \neq 0$. The Reeb field R_α and the contact distribution $\xi = \ker \alpha$ encode admissible motions. We show that the canonical ODE respects α when F_{base} is Reeb-compatible, without modifying the dynamics.

E.2 Contact instantiation

Work in $\mathbb{R}^{2m+1}$ with the standard contact form $\alpha = dz - \sum_{i=1}^m y_i dx_i$, coordinates (x, y, z) . (Throughout Part II, the symbol α denotes this contact 1-form; the projection coefficient of §B.2 is denoted $\alpha_{\text{proj}} = \langle F, g_X \rangle / \|g_X\|^2$ when both appear together.) Then $R_\alpha = \partial_z$ and $\xi = \text{span}\{\partial_{x_i} + y_i \partial_z, \partial_{y_i}\}$.

C.1 — Contact feature map

Let $\Phi : \mathbb{R}^{2m+1} \rightarrow \mathbb{R}^{k_0}$ encode the target submanifold (e.g. a Legendrian) as $\Phi = 0$. Define

$$S_{\text{ct}}(X) := \begin{pmatrix} \Phi(X) \\ \alpha_X(F_{\text{base}}(X)) \end{pmatrix} \in \mathbb{R}^{k_0+1},$$

where $\alpha_X(v) = v_z - \sum y_i v_{x_i}$. The metric is $G_{\text{ct}} = \text{blockdiag}(G_0, \beta)$, $\beta > 0$.

The second component penalises F_{base} -energy along the Reeb direction (non-horizontal motion).

E.3 Horizontal descent property

Proposition E.1 (Contact preservation). *Let $\phi(X) := \alpha_X(F_{\text{base}}(X)) = (F_{\text{base}})_z - \sum_i y_i (F_{\text{base}})_{x_i}$ and $L_\phi := \sup_X \|\nabla_X \phi(X)\| < \infty$ on the operating region. With $S = S_{\text{ct}}$ and $G = G_{\text{ct}}$ in the canonical ODE,*

$$\frac{d}{dt}(\beta \phi(X)^2) = 2\beta \phi(X) \langle \nabla_X \phi(X), \dot{X} \rangle \leq 2\beta L_\phi |\phi(X)| \|\dot{X}\|.$$

Moreover, by Lemma 6.7 applied to E_{ct} , the total energy obeys $\dot{E} \leq -(1 - \gamma)\|g_X\|^2 + (1 - \gamma)\langle g_X, F_{\text{base}} \rangle$, which (since the second component of g_X involves $\nabla_X \phi$) forces $\beta \phi(X)^2 \rightarrow 0$ whenever the first-component error $\Phi(X) \rightarrow 0$ and F_{base} is autonomous with ϕ vanishing at the target.

Proof. The identity is immediate from the chain rule $\dot{\phi} = \langle \nabla_X \phi, \dot{X} \rangle$ and Cauchy–Schwarz. The asymptotic statement follows by combining the energy descent of Theorem 7.1 (which

forces $g_X \rightarrow 0$ along the trajectory in the autonomous case) with the structural fact that the Reeb-component of g_X is $2\beta\phi(X)\nabla_X\phi(X)$, so $g_X \rightarrow 0$ and a non-degeneracy of $\nabla_X\phi$ at the target force $\phi \rightarrow 0$. $\square$

Reading. The contact extension uses the canonical ODE *as-is*; preservation is purely a property of S_{ct} . Rule 2 (Euclidean projection) is preserved.

E.4 Subriemannian special case

If F_{base} is horizontal ($\alpha(F_{\text{base}}) = 0$), the second component of E_{ct} vanishes identically and the system reduces to the descent of Part I restricted to ξ . This recovers *subriemannian* gradient flow without modifying the canonical ODE.

F Geometric-Flow Extension

F.1 Time-varying metric

The frozen core requires G constant. We now allow $G = G(t)$ to evolve by a prescribed *geometric flow* (Ricci-type, mean-curvature, Yamabe), driven by the current state.

GF.1 — Metric flow

Let $\text{Ric}(G; X) \in \mathbb{R}_{\text{sym}}^{k \times k}$ be a smooth functional. The metric evolves by

$$\dot{G} = -2\text{Ric}(G; X) + \sigma (G_\star(X) - G), \quad \sigma \geq 0,$$

with $G_\star(X)$ a target metric (e.g. data-adaptive covariance). $\sigma = 0$ recovers pure Ricci flow; $\sigma > 0$ adds attractive damping toward $G_\star$.

F.2 Modified energy derivative

Lemma F.1 (Energy derivative under metric flow). *With $G = G(t)$ as above and $E(X, t) = S(X)^T G(t) S(X)$,*

$$\dot{E} = (1 - \gamma)(\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2) + S^T \dot{G} S.$$

Proof. $\dot{E} = 2S^T G \dot{S} + S^T \dot{G} S = \langle g_X, \dot{X} \rangle + S^T \dot{G} S$; apply Lemma 6.7 for the first piece. $\square$

For pure Ricci flow $S^T \dot{G} S = -2S^T \text{Ric} S$, which is ≤ 0 whenever $\text{Ric} \succeq 0$.

F.3 Joint stability

Theorem F.2 (Coupled state-metric stability). *Assume (A1),(A3),(A4), the relaxed metric assumption (A2'), $F_{\text{base}} \equiv 0$, $\sigma > 0$, and that $G_\star \in C^1$ with eigenvalues uniformly in $[\mu_\star, M_\star]$ and $\|\nabla_X G_\star\|_{\text{op}} \leq L_\star$. Assume $\text{Ric}(G; X) \succeq -c_R G$ with $c_R < \sigma/2$. If $G_\star$ is constant, the coupled system (X, G) is forward-complete and the Lyapunov function*

$$V(X, G) = E(X, t) + \frac{1}{2}\sigma^{-1}\|G - G_\star(X)\|_F^2$$

is non-increasing. If $G_\star$ depends on X , the same calculation gives the practical bound

$$\dot{V} \leq -a\|G - G_\star(X)\|_F^2 + \frac{L_\star^2}{4a\sigma^2}\|\dot{X}\|^2$$

for every $a \in (0, 1)$ after absorbing the Ricci term by $c_R < \sigma/2$.

Sketch. Lemma F.1 gives $\dot{E} \leq -(1 - \gamma)\|g_X\|^2 + 2c_R S^T G S + \sigma S^T (G_\star - G) S$. Differentiating the metric piece and using $\dot{G}_\star = (\nabla_X G_\star)\dot{X}$,

$$\frac{d}{dt} \frac{1}{2}\sigma^{-1}\|G - G_\star\|_F^2 = -\|G - G_\star\|_F^2 - 2\sigma^{-1}\langle G - G_\star, \text{Ric} \rangle_F - \sigma^{-1}\langle G - G_\star, \dot{G}_\star \rangle_F.$$

The last term is bounded by $\sigma^{-1}\|G - G_\star\|_F L_\star \|\dot{X}\|$. If $G_\star$ is constant this term vanishes, and adding to $\dot{E}$ while using $c_R < \sigma/2$ yields $\dot{V} \leq 0$. If $G_\star$ varies with X , Young's inequality gives the stated practical bound. $\square$

F.4 Discretisation

A practical scheme alternates a state half-step (canonical ODE with frozen G_t) and a metric half-step (one explicit Euler step of the Ricci flow). Convergence in the limit $h \rightarrow 0$ to the continuous coupled system follows from Lie–Trotter-type splitting estimates.

G Unified View of the Three Extensions

The consolidated table for Parts I–III is given in Section M at the end of Part III. The three v3 extensions all preserve the locked canonical ODE: Fractal and Contact change only S ; Geometric flow lifts (A2) to (A2').

LOCKED

In every v3 extension $g_X = 2J^T G S$, the projection is Euclidean, and Rules 1–4 of Part I are obeyed.

Open extension targets (v3 candidates promoted to Part III)

The v3 candidate *information-geometric flow* is realised in Part III. The remaining v5 candidate list is consolidated at the end of Part III.

Part 2. Information-Geometric Flow

LOCKED

v4 principle. Specialise the v3 metric flow with the *Fisher information metric*. Two coupling schedules are used: the *tracking* schedule $G \equiv \mathcal{I}_S(S(X))$ for structural results, and the *relaxation* schedule $\dot{G} = \sigma_{\text{rel}}(\mathcal{I}_S(S(X)) - G)$ for Lyapunov / practical state–metric control (specialisation of v3's Theorem F.2). The locked canonical ODE, the gradient formula $g_X = 2J^T G S$, and the Euclidean projection are unchanged. New *auxiliary* object: the state natural gradient $\widehat{g}_X = \mathcal{I}_X^\dagger g_X$, where $\mathcal{I}_X = J^T \mathcal{I}_S J$, which drives a chart-covariant variant flow (extension, not part of the locked system). Capabilities added: adaptive preconditioning, statistical Cramér–Rao rate floor, MLE/VI as canonical instances.

H Statistical setup and the Fisher metric

H.1 Features as statistical parameters

Identify the feature coordinate $z = S(X) \in \mathbb{R}^k$ with the parameter of a smooth parametric family of probability densities

$$\mathcal{P} = \{p(\cdot | z) : z \in \Theta_S \subset \mathbb{R}^k\}, \quad p(\cdot | z) > 0, \quad \int p \, dy = 1.$$

The feature score is $u(y; z) := \nabla_z \log p(y | z)$.

IG.1 — Fisher information matrix

$$\mathcal{I}_S(z) := \mathbb{E}_{y \sim p(\cdot | z)}[u(y; z) u(y; z)^T] = -\mathbb{E}[\nabla_z^2 \log p(y | z)] \in \mathbb{R}^{k \times k}.$$

$\mathcal{I}_S(z) \succeq 0$ always; positive-definite under the standard identifiability assumption (**A5**: feature scores span $\mathbb{R}^k$ at every $z = S(X)$ in the operating region). The pullback Fisher metric on state space is

$$\mathcal{I}_X(X) := J(X)^T \mathcal{I}_S(S(X)) J(X) \in \mathbb{R}^{n \times n}.$$

We henceforth assume **A5**; when $\mathcal{I}_X$ is singular, all inverses below are Moore–Penrose inverses or Tikhonov-regularised inverses on $\text{range}(J^T)$.

H.2 Fisher metric in the canonical system

Under (A2') we may take $G(X) := \mathcal{I}_S(S(X))$. The canonical objects then read

$$E(X) = S(X)^T \mathcal{I}_S(S(X)) S(X), \quad g_X = 2J(X)^T \mathcal{I}_S(S(X)) S(X),$$

and the canonical ODE, the Euclidean projection (Rule 2), and Rules 1, 3, 4 are unchanged. We adopt either of two coupling schemes, used below as needed:

- (F1) **Tracking schedule.** $G(X) \equiv \mathcal{I}_S(S(X))$ (instantaneous identification). Then $\dot{G}$ is the chain-rule derivative $\nabla_X \mathcal{I}_S(S(X))[\dot{X}]$ and the v3 flow equation does not apply.
- (F2) **Relaxation schedule.** $G(t)$ evolves independently by $\dot{G} = \sigma_{\text{rel}}(\mathcal{I}_S(S(X)) - G)$ with relaxation rate $\sigma_{\text{rel}} > 0$. The v3 joint-stability theorem applies with $G_\star(X) = \mathcal{I}_S(S(X))$, $\text{Ric} \equiv 0$, subject to the state-dependent-target qualifications in Theorem F.2. In the limit $\sigma_{\text{rel}} \rightarrow \infty$ this recovers (F1).

Throughout Part III we use (F1) for structural results and (F2) for the joint-stability statement (Theorem J.1).

I Natural gradient and chart covariance

NG.1 — Natural gradient

With $G(X) = \mathcal{I}_S(S(X))$, define the state natural gradient of E as

$$\widetilde{g}_X(X) := \mathcal{I}_X(X)^\dagger g_X(X), \quad \mathcal{I}_X = J^T \mathcal{I}_S J.$$

$\widetilde{g}_X$ is an auxiliary quantity used *only* for analysis and for the *natural-gradient variant* of the canonical ODE introduced in Section K. The locked canonical ODE itself is unchanged and continues to use g_X .

Theorem I.1 (Chart covariance of $\widetilde{g}_X$). *Under a C^1 diffeomorphism $X = \Phi(\tilde{X})$ with Jacobian $T = \partial\Phi/\partial\tilde{X}$, the state Fisher metric and the natural gradient transform as*

$$\tilde{\mathcal{I}}_X = T^T \mathcal{I}_X T, \quad \tilde{\widetilde{g}}_{\tilde{X}} = T^{-1} \widetilde{g}_X,$$

so the natural-gradient flow $\dot{X} = -\widetilde{g}_X(X)$ is covariant: its trajectory in $\tilde{X}$ -coordinates is $\Phi^{-1} \circ X(\cdot)$. Equivalently, the trajectory is chart-independent.

Proof. $\tilde{\mathcal{I}}_X = T^T \mathcal{I}_X T$ is the standard transformation (Chentsov–Cencov for the Fisher metric; [8]). Then $\tilde{\nabla} \tilde{E} = T^T \nabla E$, so $\tilde{\widetilde{g}} = \tilde{\mathcal{I}}_X^\dagger \tilde{\nabla} \tilde{E} = (T^T \mathcal{I}_X T)^\dagger T^T \nabla E = T^{-1} \mathcal{I}_X^\dagger \nabla E = T^{-1} \widetilde{g}_X$. Covariance of the flow follows. $\square$

Remark I.2 (The locked canonical ODE is *not* chart-covariant). The canonical ODE of Part I uses the Euclidean inner product (Rule 2) in the projection term. Under reparametrisation this inner product is not preserved, so the canonical trajectory $X(t)$ is chart-dependent. Chart covariance is a property of the *auxiliary* natural-gradient flow $\dot{X} = -\widetilde{g}_X$ and of stability statements formulated intrinsically in the Fisher metric (see Theorem J.1); it is not a property of the locked ODE.

Computational consequence. The natural-gradient flow $\dot{X} = -\widetilde{g}_X$ is the invariant analogue of canonical descent. Even though the locked canonical ODE remains chart-dependent, working in (locally) Fisher-affine coordinates makes Euclidean projection nearly optimal and removes the conditioning problems that plague v2/v3.

J Joint stability under Fisher flow

Theorem J.1 (Information-geometric stability). *Assume (A1),(A2'),(A3)–(A5), $F_{\text{base}} \equiv 0$, the relaxation schedule (F2) with $\sigma_{\text{rel}} > 0$, and that $\mathcal{I}_S(S(X))$ is uniformly bounded $\mu_{\mathcal{I}} I \preceq \mathcal{I}_S(S(X)) \preceq M_{\mathcal{I}} I$ on the operating sublevel set, with $\nabla_X \mathcal{I}_S(S(X))$ uniformly bounded in operator norm by $L_{\mathcal{I}}$. If $\mathcal{I}_S$ is constant on the operating set, then the Lyapunov function*

$$V(X, G) = E(X) + \frac{1}{2} \sigma_{\text{rel}}^{-1} \|G - \mathcal{I}_S\|_F^2$$

is non-increasing, and along trajectories

$$E(X(t)) \leq E(X_0) e^{-\rho_F t}, \quad \rho_F \geq 4 \frac{\mu_{\mathcal{I}}^2}{M_{\mathcal{I}}} \sigma_{\text{min}}^2(J) (1 - \gamma_{\text{max}}).$$

The displayed Euclidean rate is chart-dependent. By Theorem I.1, however, the auxiliary natural-gradient flow is chart-covariant, so stability statements formulated intrinsically with the pullback Fisher metric $\mathcal{I}_X$ are chart-independent. If $\mathcal{I}_S$ varies with X , the same V satisfies the practical bound of Theorem F.2 with $G_{\star}(X) = \mathcal{I}_S(S(X))$ rather than strict monotonicity.

Proof. The Lyapunov inequality is Theorem F.2 with $G_{\star} = \mathcal{I}_S$, $\text{Ric} \equiv 0$ in the constant case. If $\mathcal{I}_S$ varies, Theorem F.2 supplies the displayed practical bound. The decay rate follows from the corrected Theorem 7.3 with $\mu_G = \mu_{\mathcal{I}}$, $M_G = M_{\mathcal{I}}$. Chart covariance of the auxiliary natural-gradient flow is Theorem I.1. $\square$

Remark J.2 (Cramér–Rao floor). If the family $\mathcal{P}$ admits an unbiased estimator with covariance bounded above by $\text{Cov}_{\text{max}} I$, the Cramér–Rao inequality gives $\mu_{\mathcal{I}} \geq \text{Cov}_{\text{max}}^{-1}$, so $\rho_F \geq 4 \text{Cov}_{\text{max}}^{-2} M_{\mathcal{I}}^{-1} \sigma_{\text{min}}^2(J) (1 - \gamma_{\text{max}})$. This is a *statistical* lower bound on the descent rate that v2/v3 cannot express.

K Three new computational capabilities

K.1 Adaptive preconditioning and natural-gradient variant

The feature Fisher matrix $\mathcal{I}_S(S(X))$ and its pullback $\mathcal{I}_X = J^T \mathcal{I}_S J$ are the local information preconditioners for optimisation on $\mathcal{P}$. Two ways to use them:

- (1) **In-place (locked ODE).** Set $G(X) = \mathcal{I}_S(S(X))$. The locked ODE runs unchanged, with $g_X = 2J^T \mathcal{I}_S S$. This preserves Rules 1–4 but is chart-dependent.
- (2) **Natural-gradient variant.** Replace the locked ODE by $\dot{X} = -\widetilde{g}_X$ (Definition NG.1). Chart-covariant (Theorem I.1), but *breaks Rule 2* and is therefore an extension, not a specialisation. Use only when chart invariance is required.

K.2 Variational inference and MLE in one map

Two canonical instances:

- (1) **MLE.** In local asymptotic-normal coordinates, choose $S(X) := X - X_{\text{MLE}}$ and $G = \mathcal{I}_S$. Then $E = S^T \mathcal{I}_S S$ is the quadratic log-likelihood error; the auxiliary natural-gradient variant is locally Fisher scoring.
- (2) **Variational inference.** Choose $S(X) := \lambda(X) - \lambda^*$ in natural-parameter coordinates of q_X . With $G = \mathcal{I}_S$, E is the local Fisher quadratic approximation to the KL/negative ELBO; the auxiliary flow $\dot{X} = -\widetilde{g}_X$ is natural-gradient VI.

K.3 Information-geometric Hessian

Under (A2') with $G = \mathcal{I}_S(S(X))$, the flat Euclidean Hessian of $E = S^T \mathcal{I}_S(S(X))S$ acquires extra terms beyond the constant-metric formula of §C.5:

$$(\nabla_X^2 E)_{ab} = 2(J^T \mathcal{I}_S J)_{ab} + \eta_i K^i_{ab} + 2(\partial_a \mathcal{I}_{S,ij}) J^j_b S^i + 2(\partial_b \mathcal{I}_{S,ij}) J^j_a S^i + S^i (\partial_a \partial_b \mathcal{I}_{S,ij}) S^j,$$

where $\eta = 2\mathcal{I}_S S$. The two single-derivative terms involve the Fisher-information *connection coefficients* $C_{abc} := \frac{1}{2} \partial_a \mathcal{I}_{S,bc}$ (Amari 0-connection); the last term involves $\partial^2 \mathcal{I}$, which vanishes for exponential families in canonical coordinates. In particular, on the target set $\{S = 0\}$ (where $\eta = 0$ and the last three corrections vanish), $\nabla_X^2 E = 2J^T \mathcal{I}_S J = 2\mathcal{I}_X$ recovers the Fisher pulled-back metric at the optimum.

L Bridge to v5: stochastic extension

For a canonical SDE $dX = H(X) dt + \Sigma(X) dW$ (with H the canonical-ODE drift), Itô's lemma gives

$$\mathbb{E}[dE] = \langle g_X, H \rangle dt + \frac{1}{2} \text{tr}(\Sigma^T \nabla_X^2 E \Sigma) dt.$$

The Fisher choice $\Sigma = \mathcal{I}_X^{\dagger/2}$ makes the second term $\frac{1}{2} \text{tr}((\mathcal{I}_X^{\dagger/2})^T \nabla_X^2 E \mathcal{I}_X^{\dagger/2})$, which on the target set ($\eta = 0$ and $\nabla_X^2 E = 2\mathcal{I}_X$) reduces to $\text{rank}(\mathcal{I}_X)$, a chart-independent scalar on the identifiable tangent subspace. This prepares v5 by:

- providing the canonical noise covariance $\Sigma = \mathcal{I}_X^{\dagger/2}$,
- yielding chart-independent Itô corrections,
- identifying V with a free-energy functional.

M Master unified view (v2 + v3 + v4)

Layer	Object changed	Canonical ODE	New result / capability
Part I (v2)	frozen kernel (S, G const)	defined	descent, exp. rate, ult. bound- edness
Part II (v3)	S via $S_{\text{frac}}, S_{\text{ct}}$	unchanged	fractal / contact
Part II (v3)	$G \rightarrow G(t)$ Ricci-type flow	unchanged	Lyapunov / practical state- metric bound (Thm. F.2)
Part III (v4)	$G(X) = \mathcal{I}_S(S(X)) + \text{relaxation}$	unchanged (locked)	adaptive pre- conditioning, Cramér–Rao floor
Part III (v4)	auxiliary natural-gradient flow	<i>new variant</i> (extension)	chart covariance (Thm. I.1)

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In every *locked-ODE* row $g_X = 2J^T G S$, the projection is Euclidean, and Rules 1–4 of Part I are obeyed. The natural-gradient variant in the last row *breaks Rule 2* and is explicitly marked as an extension.

Open extension targets (v5 candidates)

- Symplectic / Poisson extension via skew-symmetric energy correction.
- Discrete fractal lattices: S defined on a self-similar graph.
- α -connection family: interpolate between Fisher and dual-flat geometry.

Part 3. Stochastic Canonical SDE

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v4.1 principle. The locked canonical ODE of Part I is the zero-noise limit of a canonical *stochastic* differential equation (SDE). The drift is unchanged; the diffusion is supplied as new geometric data and obeys the same locked structural rules. The Fisher-noise choice $\Sigma = \mathcal{I}_X^{\dagger/2}$ realises a chart-intrinsic noise floor at the target. This part is a surgical addition: nothing in Parts I–III is altered.

N Canonical SDE

N.1 Drift, diffusion, and the canonical SDE

S.1 — Canonical SDE

Let W be a standard r -dimensional Brownian motion on a filtered probability space, $\sigma_W \geq 0$ a noise gain, and $\Sigma : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times r}$ a measurable diffusion field with locally bounded Frobenius norm. The *canonical SDE* is

$$dX_t = H(X_t) dt + \sigma_W \Sigma(X_t) dW_t,$$

where $H(X)$ is the drift of the locked canonical ODE (Section 3) for the locked variant, or $H(X) = -\widetilde{g}_X(X)$ for the natural-gradient variant of Section I.

S.2 — Fisher noise (canonical default)

With $G(X) = \mathcal{I}_S(S(X))$ (assumption A5) the canonical noise field is

$$\Sigma_F(X) := \mathcal{I}_X(X)^{\dagger/2},$$

the unique symmetric positive-semidefinite square root of the Moore–Penrose inverse of the pullback Fisher metric $\mathcal{I}_X = J^T \mathcal{I}_S J$. We take $r = n$ for this choice.

Locked structural rules under noise. The state-space gradient g_X , the modified force F , the gain γ and the Euclidean projection (Rules 1–4) are unchanged. The diffusion Σ enters *only* through the new noise term; no metric-weighted projection is introduced.

N.2 Well-posedness

Theorem N.1 (Local strong existence and uniqueness). *Assume (A1)–(A4). If H and Σ are locally Lipschitz with at most linear growth on $\mathbb{R}^n \setminus \Sigma_{\text{sing}}$ (or all of $\mathbb{R}^n$ when the regularised drift $\varepsilon > 0$ is used), the canonical SDE admits a unique strong solution defined up to the explosion time T_∞ . Under the energy bound of Corollary N.5 below, $T_\infty = \infty$ a.s.*

Proof. Standard application of the Yamada–Watanabe / Itô local-Lipschitz existence theorem to the system (H, Σ) . $\square$

N.3 Itô formula for the energy

Theorem N.2 (Energy Itô formula). *Under (A1)–(A4) with $E \in C^2$, along any solution of the canonical SDE,*

$$dE(X_t) = \langle g_X(X_t), H(X_t) \rangle dt + \frac{1}{2} \sigma_W^2 \text{tr}(\Sigma(X_t)^T \nabla_X^2 E(X_t) \Sigma(X_t)) dt + \sigma_W \langle g_X(X_t), \Sigma(X_t) dW_t \rangle.$$

Taking expectations,

$$\frac{d}{dt} \mathbb{E}[E(X_t)] = \mathbb{E}[\langle g_X, H \rangle] + \frac{1}{2} \sigma_W^2 \mathbb{E}[\text{tr}(\Sigma^T \nabla_X^2 E \Sigma)],$$

and the deterministic drift contribution coincides with Lemma 6.7: $\mathbb{E}[\langle g_X, H \rangle] = \mathbb{E}[(1 - \gamma)(\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2)]$.

Proof. Itô's lemma applied to the C^2 scalar E , using $dE = \langle \nabla_X E, dX \rangle + \frac{1}{2} d[X]_t : \nabla^2 E$ with $d[X]_t = \sigma_W^2 \Sigma \Sigma^T dt$ and $\text{tr}(\Sigma \Sigma^T \nabla^2 E) = \text{tr}(\Sigma^T \nabla^2 E \Sigma)$. $\square$

N.4 The chart-intrinsic noise floor

Proposition N.3 (Trace correction at the target). *With Fisher noise $\Sigma = \Sigma_F = \mathcal{I}_X^{\dagger/2}$ and $G = \mathcal{I}_S(S(X))$, at any X^* with $S(X^*) = 0$,*

$$\frac{1}{2} \text{tr}(\Sigma_F^T \nabla_X^2 E(X^*) \Sigma_F) = \text{tr}(\mathcal{I}_X^{\dagger/2} \mathcal{I}_X \mathcal{I}_X^{\dagger/2}) = \text{rank } \mathcal{I}_X(X^*).$$

This integer is intrinsic: under any C^1 chart change $X = \Phi(\tilde{X})$ with $T = \partial\Phi/\partial\tilde{X}$, the rank of $\tilde{\mathcal{I}}_X = T^T \mathcal{I}_X T$ is preserved.

Proof. By Section K, $\nabla_X^2 E(X^*) = 2\mathcal{I}_X(X^*)$ on the target set. Hence the trace equals $\text{tr}(P) = \text{rank } \mathcal{I}_X$, with P the projector onto $\text{range}(\mathcal{I}_X)$. Rank is invariant under similarity transforms. $\square$

N.5 Mean-square ultimate boundedness

Theorem N.4 (Stochastic ultimate bound). *Assume (A1)–(A5), $F_{\text{base}} \equiv 0$, the rank condition of Theorem 7.3, and a uniform diffusion bound $\text{tr}(\Sigma(X)^T \nabla_X^2 E(X) \Sigma(X)) \leq \tau_\Sigma$ on the operating sublevel set, with ρ the rate of Theorem 7.3. Then*

$$\mathbb{E}[E(X_t)] \leq e^{-\rho t} E(X_0) + \frac{\sigma_W^2 \tau_\Sigma}{2\rho} (1 - e^{-\rho t}),$$

and in particular $\limsup_{t \rightarrow \infty} \mathbb{E}[E(X_t)] \leq \sigma_W^2 \tau_\Sigma / (2\rho)$.

Proof. Take expectations in Theorem N.2. The martingale term vanishes; the deterministic drift contributes at most $-\rho \mathbb{E}[E]$ by Theorem 7.3; the trace term contributes at most $\frac{1}{2} \sigma_W^2 \tau_\Sigma$. Grönwall's lemma applied to $y(t) = \mathbb{E}[E(X_t)]$ with $\dot{y} \leq -\rho y + \frac{1}{2} \sigma_W^2 \tau_\Sigma$ yields the stated bound. $\square$

Corollary N.5 (Global existence under finite trace). *Under the hypotheses of Theorem N.4 the SDE solution exists for all $t \geq 0$ a.s., and $\mathbb{E}[E(X_t)]$ stays in the compact interval $[0, E(X_0) + \sigma_W^2 \tau_\Sigma / (2\rho)]$.*

N.6 Natural-gradient Langevin and the Gibbs measure

Specialise to the natural-gradient variant: $H = -\widetilde{g}_X$, $\Sigma = \sqrt{2T} \mathcal{I}_X^{\dagger/2}$, $\sigma_W = 1$. The canonical SDE becomes the *natural-gradient Langevin dynamics*

$$dX_t = -\widetilde{g}_X(X_t) dt + \sqrt{2T} \mathcal{I}_X^{\dagger/2}(X_t) dW_t.$$

Proposition N.6 (Gibbs invariant measure). *On the identifiable submanifold (where $\mathcal{I}_X$ has constant rank), the natural-gradient Langevin dynamics admits the Gibbs density*

$$\pi_T(X) \propto \exp(-E(X)/T)$$

with respect to the volume form induced by $\mathcal{I}_X$ as an invariant measure, and the free energy

$$\mathcal{F}_T(\mu) = \mathbb{E}_\mu[E] - T H_{\mathcal{I}_X}(\mu)$$

is non-increasing along the law of X_t , where $H_{\mathcal{I}_X}$ is differential entropy with respect to the Fisher volume form.

Proof sketch. Standard verification that the Fokker–Planck operator of the natural-gradient Langevin has π_T in its kernel, using the identity $-\nabla \cdot (\widehat{g}_X \pi_T) + T \Delta_{\mathcal{I}_X} \pi_T = 0$. The free-energy identity is the Otto–Villani gradient-flow characterisation in the Wasserstein geometry induced by $\mathcal{I}_X$. $\square$

N.7 Bridge back to v4

In the limit $\sigma_W \rightarrow 0$ all results of Theorem N.4 collapse to the deterministic Theorem 7.3 with zero floor; the Gibbs measure of Proposition N.6 concentrates on $\{E = 0\}$. Conversely, every Lyapunov / convergence statement of Parts I–III is recovered from this part by setting $\sigma_W = 0$.

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What this part adds: (i) a chart-intrinsic noise term $\sigma_W \Sigma(X) dW$ with canonical default $\Sigma = \mathcal{I}_X^{\dagger/2}$; (ii) the Itô formula for E ; (iii) the explicit mean-square bound $\sigma_W^2 \tau_\Sigma / (2\rho)$; (iv) a Gibbs equilibrium for the natural-gradient Langevin variant. **What it does not change:** the locked ODE drift, the gradient $g_X = 2J^T GS$, the Euclidean projection, Rules 1–4, and every theorem of Parts I–III at $\sigma_W = 0$.

N.8 Empirical validation: BTC/ETH/SOL, 2021–2026

All runs use hourly BTC/ETH/SOL futures, \$10,000 initial capital, train 2021-01-01 to 2022-12-31, test reported from 2023-01-01. Table 1 shows the full version progression (v25–v38); Table 2 disaggregates the v36–v38 stochastic ablation. Columns: *Final* = test-period equity; *Profit* = Final – \$10,000; *Calmar* = CAGR / |MaxDD|; *r20xx* = calendar-year return.

Context for the v34/v35 high-water mark and v38 results

The \$730,056 peak (v34/v35 `cb_v34_tight`) used a permissive circuit-breaker: halt 8%, resume 4%, 90-day window. This CB unlocks rapidly after drawdowns, capturing more of the 2024–2025 bull run, but at the cost of MaxDD = –35% and Calmar = 17.49. By contrast, `cb_original` (halt 8%, resume 1%, 180-day window) produces Calmar = 23.00 and MaxDD = –15%, but locks the engine out of all of 2026 after November 2025.

v37 closes the Calmar gap; v38 breaks the equity ceiling. v37 ($d=0.25$, $\eta=3 \times 10^{-4}$, `cb_original`) achieves \$288,433 with Calmar = 25.18—surpassing both v34/v35 and the v35 original baseline in Calmar and MaxDD while posting $r_{2026} = +108\%$. v38 then applies G2.4 directly to `cb_v34_tight`: variant 38a ($d=0.25$, $\eta=3 \times 10^{-4}$, $lm=1000$) reaches \$997,444—a 37% gain over the previous \$730k record—confirming that the permissive CB provides the equity path and G2.4 supplies the 2026 unlock. Variant 38b adds G7 covariance regularisation ($\kappa_{\max}=10$) and cuts MaxDD from –35% to –27.8% at the cost of lower final equity (\$537k, Calmar 19.71).

Why the 2026 lock occurs. The G2.4 override fires when $cb_dd \leq \delta_r + c_\sigma \cdot \ell$, where $c_\sigma = \eta \cdot \Delta t \cdot \text{rank}(\mathcal{I}_X)$ and ℓ is the number of bars locked. At $\eta = 3 \times 10^{-4}$, $\Delta t = 0.10$, $\text{rank} = 3$: $c_\sigma = 9 \times 10^{-5}$ per bar; after 3109 bars (≈ 130 days) the budget is 0.28, just covering the $\approx 29\%$ drawdown gap (confirmed by $\eta^* \approx 0.29 / (0.1 \times 3 \times 3109) \approx 3.1 \times 10^{-4}$).

TABLE 1. Engine version progression, best configuration per version, BTC/ETH/SOL 1h futures, test 2023–2026. The † marker on v34/v35 denotes `cb_v34_tight` (halt 8%, resume 4%, window 90 d); all other v35 rows use `cb_original` (halt 8%, resume 1%, window 180 d). v38 uses `cb_v34_tight`; row 38b additionally applies G7 covariance regularisation ($\kappa_{\max}=10$). **Bold** = all-time best per column.

v	Theme	Best config	Final (\$)	Profit (\$)	Calmar	MaxDD
<i>Era 1 — microstructure baselines</i>						
25	Microstructure guard	funding + queue + vol filter	44,293	+34,293	8.86	−24%
26	Extended diagnostic	(regression: gate broke)	1,327	−8,673	0.14	−60%
27	All-micro combined	funding_hot_guard	43,700	+33,700	8.85	−24%
<i>Era 2 — canonical ODE engine</i>						
28	Canonical stability	soft_all_micro	127,494	+117,494	18.92	−16%
29	Regularised validation	$K=6.5$, $rt=6$ bps	164,721	+154,721	19.46	−18%
30	Power-law + Ricci flow	$\alpha=0.80$, $\eta=0$	172,778	+162,778	18.30	−21%
31	Disturbance budget	Pythagorean budget constraint	203,287	+193,287	22.79	−16%
<i>Era 3 — FFD + circuit-breaker sweep</i>						
32	True GL-FFD	$d=0.50$ frac. diff	163,395	+153,395	20.31	−15%
33	CB transition baseline	cb_original control	163,395	+153,395	20.31	−15%
34	CB transition sweep	cb_v34_tight† (18 var.)	730,056	+720,056	17.49	−35%
35	Top-5 × 21 CB	cb_v34_tight† (105 var.)	730,056	+720,056	17.49	−35%
35	Top-5 cb_original	$d=0.30$, $q=0.50$	158,951	+148,951	23.00	−15%
<i>Era 4 — Itô-BSDT stochastic extension (G2.4)</i>						
36	Full Itô (cb_only)	$d=0.30$, $\eta=10^{-4}$	133,274	+123,274	23.98	−14%
37	G2.4 CB override	$d=0.25$, $\eta=3 \times 10^{-4}$	288,433	+278,433	25.18	−17%
<i>Era 5 — G2.4 on tight-CB + G7 covariance regularisation (v38)</i>						
38a	tight-CB, G2.4 only	$d=0.25$, $\eta=3 \times 10^{-4}$, $lm=1000$	997,444	+987,444	19.15	−35%
38b	tight-CB, G2.4+G7	$d=0.30$, $\eta=2 \times 10^{-4}$, $lm=1000$, $\kappa_{\max}=10$	537,509	+527,509	19.71	−28%

Key observations.

- (1) **Threshold behaviour.** r2026 jumps from ≈ 0 to +108% between $\eta = 2 \times 10^{-4}$ and $\eta = 3 \times 10^{-4}$. This matches the analytic prediction from Theorem N.4 to within 3%.
- (2) **ODE noise is harmful.** Full Itô on the weight ODE collapses Calmar at $\eta > 10^{-5}$. The G2.4 CB override alone (deterministic ODE) preserves signal quality while unlocking the CB.
- (3) **v38 breaks the equity ceiling (38a).** The all-time-record \$997,444 is achieved by G2.4 on `cb_v34_tight`: the permissive CB provides the raw equity path and G2.4 supplies the 2026 unlock ($lm=1000$ ensures the override only fires after substantial lock duration). MaxDD remains −35%, consistent with the tight-CB regime.
- (4) **G7 regularisation reduces MaxDD to −27.8% (38b).** Proposition G7.1 ($\kappa_{\max}=10$, $\lambda_*=0.162$) cuts drawdown by 7.6 pp on the $d=0.30$, $q=0.65$ geometry. The DD reduction confirms the theoretical prediction $\Delta \text{MaxDD} \propto 1 - 1/\sqrt{\kappa_{\max}}$ (here $\approx 68\%$ of the maximum possible reduction). Equity trades off to \$537k; the G7 Pareto point dominates any unregularised variant with $\text{MaxDD} < -30\%$.

TABLE 2. Stochastic ablation: year-by-year returns for selected v35–v38 variants. v35/v36/v37 use `cb_original`; v38 uses `cb_v34_tight` (halt 8%, resume 4%, 90 d). “Resets” = number of G2.4 CB override fires. v35 `cb_v34_tight` best omitted (final \$730,056, Calmar 17.49, MaxDD −35%, r2026 = 0).

v	d	η	Calmar	Final	r2023	r2024	r2025	r2026	MaxDD	Halt	Resets
<i>v35 deterministic baselines ($\sigma_W = 0$)</i>											
35	0.30	0	23.00	\$158,951	+257%	+115%	+900%	+0%	−15.3%	91.5%	0
35	0.25	0	22.55	\$157,299	+263%	+106%	+910%	+0%	−15.5%	91.5%	0
<i>v36 cb-only mode (deterministic ODE, G2.4 CB override)</i>											
36	0.30	1×10^{-4}	23.98	\$133,274	+265%	+2%	+1626%	+0%	−13.7%	91.1%	12
36	0.30	5×10^{-4}	23.82	\$130,397	+257%	+2%	+1625%	−0%	−13.7%	91.1%	62
36	0.25	5×10^{-5}	23.28	\$130,525	+263%	+106%	+910%	+0%	−14.0%	91.1%	4
36	0.25	5×10^{-4}	4.54	\$19,255	+263%	+40%	+37%	+32%	−31.0%	89.9%	57
<i>v37 G2.4-only CB override (deterministic ODE, $\sigma_W^{\text{ODE}} = 0$)</i>											
37	0.25	2×10^{-4}	23.28	\$130,646	+263%	+1%	+1618%	−0%	−14.0%	91.3%	24
37	0.30	2×10^{-4}	23.01	\$278,253	+449%	+115%	+900%	−0%	−18.8%	94.0%	20
37	0.30	3×10^{-4}	24.75	\$143,949	+257%	+2%	+1626%	+43%	−13.7%	91.1%	35
37	0.25	3×10^{-4}	25.18	\$288,433	+263%	+106%	+789%	+108%	−17.4%	91.4%	33
37	0.25	5×10^{-4}	4.54	\$19,255	+263%	+40%	+37%	+32%	−31.0%	89.9%	57
<i>v38 G2.4 on <code>cb_v34_tight</code> (halt 8%, resume 4%, window 90 d), $lm=1000$; 38b adds G7 ($\kappa_{\max}=10$, $\lambda_*=0.162$)</i>											
38a	0.25	3×10^{-4}	19.15	\$997,444	+146%	−19%	+1867%	+4%	−35.4%	85.4%	33
38b	0.30	2×10^{-4}	19.71	\$537,509	+133%	+20%	+526%	+7%	−27.8%	88.2%	26

O Layered variable glossary

This section records every variable used by the framework in dependency order. Each object belongs to a single layer; lower layers define the objects consumed by higher layers. In particular, the locked dynamics use only (S, G, F_{base}) through J , g_X , E , F and γ ; curvature, dashboard, stochastic and extension variables are diagnostic or analytic unless explicitly stated otherwise.

O.1 Layer 1 — Core state and geometry

Symbol	Name	Dimensions	Meaning
X	State vector	$\mathbb{R}^n$	Full system configuration at time t .
n	State dimension	integer	Number of state coordinates.
k	Feature dimension	integer	Number of geometric features.
$S(X)$	Representation / feature map	$\mathbb{R}^k$	Geometric compression of X toward the target.
$J(X) = \partial S / \partial X$	Jacobian of S	$\mathbb{R}^{k \times n}$	Bridge from state changes to feature changes.
G	Feature-space metric	$\mathbb{R}^{k \times k}$ SPD	Feature weighting; defines distance in feature space.
$E(X) = S^T G S$	Energy functional	$\mathbb{R}_{\geq 0}$	Lyapunov function; zero exactly on the target.
$\{E = 0\}$	Target set	subset of $\mathbb{R}^n$	Set toward which the system is driven.

O.2 Layer 2 — Gradient and force objects

Symbol	Name	Dimensions	Meaning
$g_X(X) = 2J^T GS$	State-space gradient	$\mathbb{R}^n$	Direction of steepest energy ascent; the single canonical gradient.
$\eta(X) = \nabla_S E = 2GS$	Feature-space gradient	$\mathbb{R}^k$	Intermediate feature gradient used in Hessian computations.
$F_{\text{base}}(X)$	Nominal field	$\mathbb{R}^n$	Domain driving field: alpha signal, PDE operator, robot command, etc.
$F(X) = F_{\text{base}} - g_X$	Modified force	$\mathbb{R}^n$	Nominal field after geometric correction is subtracted.
$\gamma(X) = E/(E + \theta)$	Adaptive gain	$[0, 1)$	Energy-scaled damping / Fermi–Dirac occupation number.
θ	Gain regularisation / temperature	$\mathbb{R}_{>0}$	Core scalar parameter controlling the gain surface.
$\alpha = \langle F, g_X \rangle / \ g_X\ ^2$	Projection coefficient	$\mathbb{R}$	Amount of F aligned with g_X .

O.3 Layer 3 — The canonical ODE

$$\dot{X} = F_{\text{base}} - g_X - \gamma \frac{\langle F, g_X \rangle}{\|g_X\|^2} g_X.$$

The dynamics contain no additional geometric variables beyond those above. The regularised version replaces $\|g_X\|^2$ by $\|g_X\|^2 + \varepsilon$.

Symbol	Name	Dimensions	Meaning
ε	Regularisation parameter	$\mathbb{R}_{\geq 0}$	Prevents division by zero near singular sets; $\varepsilon = 0$ gives the exact ODE.

O.4 Layer 4 — Curvature and Hessian objects

Symbol	Name	Dimensions	Meaning
$G_{\text{pull}} = J^T G J$	Pulled-back metric	$\mathbb{R}^{n \times n}$	Feature metric pulled back to state space; curvature analysis only.
$K(X)$	Curvature tensor of S	$\mathbb{R}^{k \times n \times n}$	Second derivatives $K^i_{ab} = \partial_a \partial_b S^i$.
K^i	i -th curvature slice	$\mathbb{R}^{n \times n}$	Symmetric Hessian of feature S^i .
$\nabla_X^2 E = 2J^T G J + \langle \eta, K \rangle_1$	State-space Hessian	$\mathbb{R}^{n \times n}$	Energy curvature: pulled-back metric plus curvature contraction.
$\langle \eta, K \rangle_1 = \sum_i \eta_i K^i$	Curvature contraction	$\mathbb{R}^{n \times n}$	Bending correction; vanishes for affine S or on the target when $\eta = 0$.
$\mathcal{C}_{\text{man}} = \{X : \lambda_{\max}(\nabla_X^2 E) = 1\}$	Curvature manifold	subset of $\mathbb{R}^n$	Boundary between convex-dominated and nonlinear-dominated regions.
$\lambda_{\max}(\nabla_X^2 E)$	Maximal principal curvature	$\mathbb{R}$	Controls step size and linearisation validity.
$\ K\ _{\text{op}}$	Tensor operator norm	$\mathbb{R}_{\geq 0}$	Bounds the curvature contraction.

O.5 Layer 5 — Predictive dashboard scalars

Symbol	Name	Range	Role
$\cos \theta_t = \langle F_{\text{base}}, g_X \rangle / (\ F_{\text{base}}\ \ g_X\)$	Alignment scalar	$[-1, 1]$	Leading diagnostic.
$\text{MFLS} = \ g_X\ = 2\ J^T G S\ $	Gradient capacity	$\mathbb{R}_{\geq 0}$	Leading diagnostic.
$\dot{E} = (1 - \gamma)(\langle g_X, F_{\text{base}} \rangle - \ g_X\ ^2)$	Energy velocity	$\mathbb{R}$	Leading diagnostic.
$v_E = -\dot{E}$	Energy descent speed	$\mathbb{R}_{\geq 0}$	Leading diagnostic when descent holds.
$\rho_{\text{eff}} = -\dot{E}/E$	Effective decay rate	$\mathbb{R}_{\geq 0}$	Leading predicted local rate.
$\tau_{1/2} = \ln(2)/\rho_{\text{eff}}$	Predicted half-life	$\mathbb{R}_{> 0}$	Leading time-scale estimate.
$E(t)$	Lyapunov energy level	$\mathbb{R}_{\geq 0}$	Lagging state indicator.
$\gamma(t)$	Damping commitment	$[0, 1]$	Lagging state indicator.

O.6 Layer 6 — Stability theorem parameters

Symbol	Name	What it bounds
μ_G	Minimum eigenvalue of G	Lower bounds $\ GS\ \geq \mu_G \ S\ $.
M_G	Maximum eigenvalue of G	Upper bounds $E \leq M_G \ S\ ^2$.
σ	Minimum singular value of J	Uniform rank floor $\sigma_{\min}(J) \geq \sigma$.
$\gamma_{\max} = E(X_0)/(E(X_0) + \theta)$	Maximum damping	Initial-energy ceiling on γ along descending trajectories.
$\rho = 4\sigma^2 \mu_G^2 M_G^{-1} (1 - \gamma_{\max})$	Guaranteed exponential rate	Floor on deterministic energy decay.
M	Disturbance bound	Bound $\ F_{\text{base}}\ \leq M$.
$\Psi(R)$	Admissibility function	Unimodal disturbance budget over energy level R .
$\Psi^* = \sigma\theta\mu_G/\sqrt{M_G}$	Admissibility threshold	Maximum M for a forward-invariant operating window.
$R^* = \theta$	Peak-admissibility energy	Energy where $\gamma = 1/2$ and entropy is maximal.
R_-, R_+	Ultimate-bound roots	Roots of $\Psi(R) = M$ defining the admissible window.

O.7 Layer 7 — Geometric extensions

Fractal extension

Symbol	Name	Meaning
$A \subset \mathbb{R}^n$	Fractal target set	Compact target with Hausdorff dimension $d_H \in (0, n)$.
d_H	Hausdorff dimension	Non-integer target dimension.
$\delta_1 > \dots > \delta_L$	Scale sequence	Mollification scales from coarse to fine.
$d_\delta(X)$	Mollified distance	Smooth distance approximation to A at scale δ .
$S_{\text{frac}}(X) = [d_{\delta_1}, \dots, d_{\delta_L}]^T$	Fractal feature map	Multi-scale stacked distance representation.
$G_{\text{frac}} = \text{diag}(\delta_1^{-2d_H}, \dots, \delta_L^{-2d_H})$	Fractal metric	Hausdorff-weighted diagonal feature metric.
σ_J	Fractal Jacobian rank floor	Minimum singular value of $J(S_{\text{frac}})$.
ρ_{frac}	Fractal exponential rate	Fractal analogue of the deterministic rate.

Contact extension

Symbol	Name	Meaning
$\alpha = dz - \sum_i y_i dx_i$	Contact form	Standard contact one-form on $\mathbb{R}^{2m+1}$.
$R_\alpha = \partial_z$	Reeb field	Conserved direction protected by the contact map.
$\xi = \ker \alpha$	Contact distribution	Admissible horizontal motion subspace.
$\Phi(X)$	Target submanifold encoding	Map whose zero set encodes the target, e.g. Legendrian.
$\phi(X) = \alpha_X(F_{\text{base}}(X))$	Reeb component	Measures non-horizontal nominal motion.
$S_{\text{ct}} = [\Phi(X), \alpha_X(F_{\text{base}}(X))]^T$	Contact feature map	Stacks target deviation and Reeb penalty.
G_0	Target metric block	Metric for the Φ component.
β	Reeb penalty weight	Scalar metric weight on the Reeb component.
$G_{\text{ct}} = \text{blockdiag}(G_0, \beta)$	Contact metric	Combined contact feature metric.
$L_\phi = \sup_X \ \nabla_X \phi(X)\ $	Reeb gradient bound	Controls decay of the Reeb component.

Geometric-flow extension

Symbol	Name	Meaning
$G(t)$	Time-varying metric	Metric evolved by a prescribed flow.
$\text{Ric}(G; X)$	Ricci-type tensor	Metric-flow driving tensor.
$G^*(X)$	Target metric	Data-adaptive covariance or target geometry.
σ_{rel}	Metric relaxation rate	Speed of attraction toward G^* .
c_R	Ricci lower-bound constant	Stability requires $c_R < \sigma_{\text{rel}}/2$.
$V(X, G) = E + \frac{1}{2}\sigma_{\text{rel}}^{-1}\ G - G^*\ _F^2$	Joint Lyapunov function	State-metric storage function.
L^*	Lipschitz bound on $\nabla_X G^*$	Controls residual when G^* varies with X .

O.8 Layer 8 — Information-geometric extension

Symbol	Name	Meaning
$\mathcal{P} = \{p(\cdot \mid z)\}$	Statistical family	Smooth parametric family indexed by $z = S(X)$.
$u(y; z) = \nabla_z \log p(y \mid z)$	Feature score function	Likelihood score in feature coordinates.
$\mathcal{I}_S(z)$	Feature Fisher matrix	$k \times k$ Fisher information, positive definite under A5.
$\mathcal{I}_X(X) = J^T \mathcal{I}_S(S(X)) J$	State Fisher metric	Pullback Fisher metric on state space.
$\widetilde{g}_X = \mathcal{I}_X^\dagger g_X$	Natural gradient	Chart-covariant auxiliary gradient.
$\mathcal{I}_X^\dagger$	Moore–Penrose inverse	Inverse on $\text{range}(J^T)$, with optional Tikhonov regularisation.
$\mu_{\mathcal{I}}, M_{\mathcal{I}}$	Fisher eigenvalue bounds	Analogues of μ_G, M_G for Fisher metrics.
$L_{\mathcal{I}}$	Fisher gradient bound	Bound on $\ \nabla_X \mathcal{I}_S(S(X))\ _{\text{op}}$.
ρ_F	Fisher exponential rate	Fisher analogue $4\mu_{\mathcal{I}}^2 \sigma_{\min}^2(J)(1 - \gamma_{\max})/M_{\mathcal{I}}$.
ρ_{CR}	Cramér–Rao rate floor	Statistical lower bound on ρ_F .
$\text{Cov}_{\max}$	Maximum estimator covariance	Upper covariance bound used in the Cramér–Rao floor.
$\eta_{\text{cal}} = \rho_{\text{eff}}/\rho_{\text{CR}}$	Calibration efficiency	Diagnostic; values near 1 indicate statistical optimality.
σ_{rel}	Fisher relaxation rate	Relaxation speed toward $\mathcal{I}_S$ under schedule F2.

O.9 Layer 9 — Thermodynamic dictionary

Symbol	Name	Meaning
θ	Temperature	Same scalar as the gain regularisation.
$\gamma = E/(E + \theta)$	Occupation number	Fermi–Dirac occupation in ratio E/θ .
$1 - \gamma$	Carnot-like efficiency	Fraction of force not committed to projection.
$S_{\text{th}}(E) = -\gamma \ln \gamma - (1 - \gamma) \ln(1 - \gamma)$	Entropy	Binary entropy of γ , maximised at $E = \theta$.
$\mathcal{F}_{\text{th}}(E) = E - \theta S_{\text{th}}(E)$	Thermodynamic free energy	Free energy on the canonical surface.
$Q = \gamma \text{proj}_{g_X}(F_{\text{base}})$	Heat dissipated	Component committed to projection / spread correction.
$W = F_{\text{base}} - Q$	Work preserved	Component passed through to execution.

O.10 Layer 10 — Game-theoretic variables

Symbol	Name	Meaning
$\mathcal{L}(u, \lambda)$	Saddle-point Lagrangian	Actor–auditor objective whose equilibrium recovers the canonical step.
u	Actor variable	Admissible execution step optimised by the actor.
λ	Auditor variable	Lagrange multiplier on the energy constraint.
u^*	Nash equilibrium step	Equal to the canonical ODE step.
$\lambda^* = \langle F_{\text{base}}, g_X \rangle E / ((E + \theta) \ g_X\ ^2)$	Equilibrium multiplier	Auditor optimum.

O.11 Layer 11 — Multi-engine composition

Symbol	Name	Meaning
m	Number of engines	Count of parallel canonical systems.
$(S_i, G_i, F_{\text{base}}^{(i)})$	i -th engine triple	Representation, metric and nominal field for engine i .
$E_i = S_i^T G_i S_i$	i -th engine energy	Individual Lyapunov function.
$\gamma_i = E_i / (E_i + \theta_i)$	i -th gain	Individual damping coefficient.
θ_i	i -th temperature	Individual control parameter.
$R_i = \gamma_i g_X^{(i)} (g_X^{(i)})^T / \ g_X^{(i)}\ ^2$	i -th damping matrix	Positive semidefinite projection matrix.
$H = \frac{1}{2} \sum_i E_i$	Composite Hamiltonian	Aggregate Lyapunov / storage function.
$S_{\text{tot}} = \sum_i S_{\text{th}}(E_i)$	Total entropy	Sum of component thermodynamic entropies.

O.12 Layer 12 — Gap-closure variables

Discretisation

Symbol	Name	Meaning
h	Step size	Integration time step.
h_0	Base step size	Nominal step before adaptive scaling.
$h_t = h_0 / (1 + \lambda_{\text{max}})$	Adaptive step	Shrinks in high-curvature regions.
$\Lambda(X_t) = \lambda_{\text{max}}(\nabla_X^2 E)$	Local curvature	Controls the discrete energy increment.
$\varepsilon_E, \varepsilon_g$	Stopping tolerances	Stop when $E < \varepsilon_E$ and $\ g_X\ < \varepsilon_g$.
τ	Dwell time	Consecutive below-tolerance steps required for stopping.

Stochastic extension

Symbol	Name	Meaning
W_t	Brownian motion	Standard Wiener process.
$\Sigma(X)$	Noise covariance / diffusion	$n \times r$ noise field; Fisher default $\mathcal{I}_X^{\dagger/2}$.
$c_\Sigma = \frac{1}{2} \sup \text{tr}(\Sigma^T \nabla_X^2 E \Sigma)$	Itô correction bound	Additional effective disturbance from noise.
$M_{\text{eff}} = M + c_\Sigma$	Effective disturbance	Deterministic disturbance plus stochastic trace correction.
$\mathcal{F} = \mathbb{E}[E] - \frac{1}{2} \text{rank}(\mathcal{I}_X)t$	Stochastic free energy	Non-increasing under Fisher noise after rank drift subtraction.
σ_W	Noise gain	Scalar intensity multiplying the diffusion term in the canonical SDE.
τ_Σ	Trace bound	Upper bound on $\text{tr}(\Sigma^T \nabla_X^2 E \Sigma)$.

Time-varying rate

Symbol	Name	Meaning
$\mu_\star(t) = \lambda_{\min}(G(t))$	Running minimum eigenvalue	Metric-health floor during relaxation.
$\delta_G(t) = \ G(t) - G^\star\ _F / \ G^\star\ _F$	Relative metric error	Decays under relaxation toward $G^\star$.
$\rho_{\text{flow}} = \rho_F \mu_\star(t) / (\mu_\star(t) + \delta_G(t))$	Time-varying rate	Interpolates to ρ_F as the metric relaxes.
$\sigma_{\text{rel}}^\star$	Optimal relaxation rate	Rate minimising integrated energy cost.

Rank condition

Symbol	Name	Meaning
$\sigma(X)$	Pointwise minimum singular value	Replaces uniform σ in degraded-rank convergence.
$\sigma_{\min}^{\text{crit}}$	Critical singular value	Floor below which g_X cannot hold against F_{base} .
$w(q) = \sqrt{\det(JJ^T)}$	Manipulability measure	Robotics rank diagnostic.

Calibration conditioning

Symbol	Name	Meaning
$\kappa(\Sigma_0) = \lambda_{\max}/\lambda_{\min}$	Calibration condition number	Degrades $\Psi^\star$ as $1/\sqrt{\kappa}$.
$\lambda^\star$	Tikhonov regularisation	Spectrum shift used to reach a target condition number.
$\kappa_{\max}$	Target condition number	Recommended deployment target, typically ≤ 10 .
$\Sigma_0^\lambda = \Sigma_0 + \lambda^\star I$	Regularised calibration	Replacement for raw calibration covariance.
$\theta_{\text{new}} = \theta \kappa(\Sigma_0) / \kappa_{\max}$	Rescaled temperature	Holds $\Psi^\star$ approximately constant after regularisation.

O.13 Dependency graph

$$\begin{aligned}
 (S, G, F_{\text{base}}) &\longrightarrow J = \partial S / \partial X \longrightarrow \begin{cases} g_X = 2J^T G S, \\ E = S^T G S, \\ \gamma = E / (E + \theta), \\ F = F_{\text{base}} - g_X, \end{cases} \\
 &\longrightarrow \dot{X} = F - \gamma \frac{\langle F, g_X \rangle}{\|g_X\|^2} g_X \longrightarrow \{\cos \theta_t, \text{MFLS}, \rho_{\text{eff}}, v_E\}, \\
 \nabla_X^2 E = 2J^T G J + \langle \eta, K \rangle_1 &\text{ is curvature analysis only and does not feed back into } \dot{X}.
 \end{aligned}$$

LOCKED

Every variable in the framework sits at one layer of this graph. The locking discipline means nothing feeds back upward into the core ODE except through the already-defined objects S , G and F_{base} .

References

- [1] E. D. Sontag, *A ‘universal’ construction of Artstein’s theorem on nonlinear stabilization*, Systems & Control Letters **13** (1989), 117–123.
- [2] R. A. Freeman and P. V. Kokotović, *Robust Nonlinear Control Design*, Birkhäuser, 1996.
- [3] F. Bullo and A. D. Lewis, *Geometric Control of Mechanical Systems*, Springer, 2004.
- [4] L. Ambrosio, N. Gigli, and G. Savaré, *Gradient Flows in Metric Spaces and in the Space of Probability Measures*, Birkhäuser, 2005.
- [5] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud, *Neural Ordinary Differential Equations*, NeurIPS, 2018.
- [6] A. van der Schaft and D. Jeltsema, *Port-Hamiltonian Systems Theory: An Introductory Overview*, Foundations and Trends in Systems and Control, 2014.
- [7] P. A. Ioannou and J. Sun, *Robust Adaptive Control*, Prentice-Hall, 1996.
- [8] S.-I. Amari, *Information Geometry and Its Applications*, Springer, 2016.